▶ Notebook

Lecture 5 at a glance: eight parts, numbered sections

Opening: noise to a digit

0.1 Noise to a digit ▶
0.2 Three generators ▶
0.3 The idea ▶

Part 1 · Destroying is easy

1.1 One noising step ▶
1.2 A toy schedule, by hand ▶
1.3 The shortcut ▶
1.4 The real schedule ▶
1.5 Watch our 3 dissolve ▶

Part 2 · Undoing it: one step back

2.1 What we want ▶
2.2 Big jumps back are hard, small ones easy ▶
2.3 Guess the noise ▶
2.4 One step back ▶

Part 3 · Training: labels for free

3.1 The training example ▶
3.2 Why the network needs t ▶
3.3 The loss ▶
3.4 The network ▶
3.5 Training, live ▶
3.6 What the network believes ▶

Part 4 · Generating: walk back from noise

4.1 One step back, by hand ▶
4.2 The toy, all the way back ▶
4.3 The loop ▶
4.4 Watch it ▶
4.5 Why the fresh noise? ▶
4.6 New digits ▶

Part 5 · Steering: labels, guidance and text

5.1 A 3 on demand ▶
5.2 Classifier-free guidance ▶
5.3 The guidance dial ▶
5.4 Latent diffusion ▶
5.5 Text in ▶

Part 6 · Faster sampling

6.1 DDIM: fewer steps ▶
6.2 The schedule matters ▶

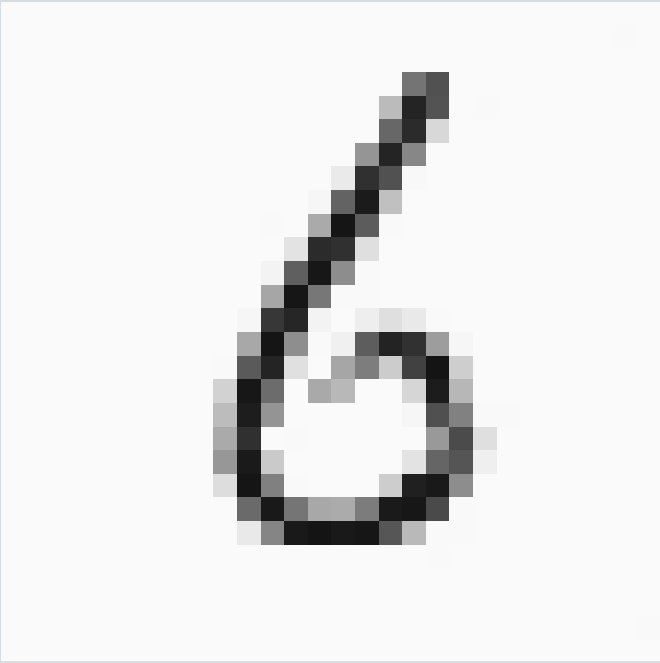
Part 7 · What the network learns, and wrap-up

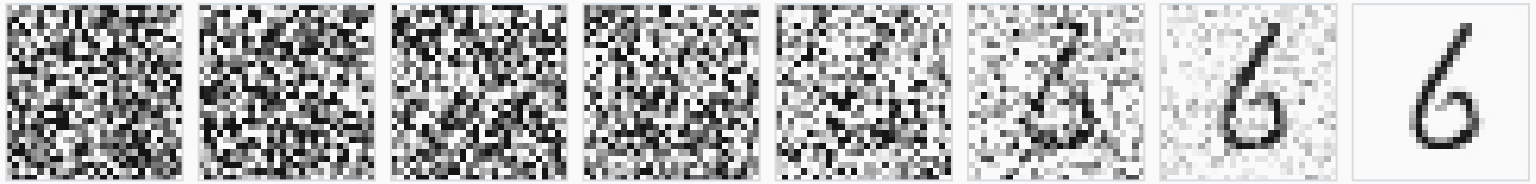
7.1 Arrows to the data ▶
7.2 It is a VAE ▶
7.3 Does it just copy? ▶
7.4 Wrap-up ▶

> Two sessions. Parts 0–4 build a working diffusion model, following every number by hand. Parts 5–7 steer it with labels and text, make it faster, and show what it learns. The ▶ Notebook link on every slide opens its section in Colab.

▶ Notebook 0.1

Static in, a handwritten digit out





t = 1000 t = 850 t = 700 t = 550 t = 400 t = 250 t = 100 t = 0

step 1000 of 1,000

a digit that is not in the training set

our network: 814,145 parameters, trained on MNIST

> …and a thousand small steps later there is a digit, which is not in the training set. Today: how, and why it works.

▶ Notebook 0.1

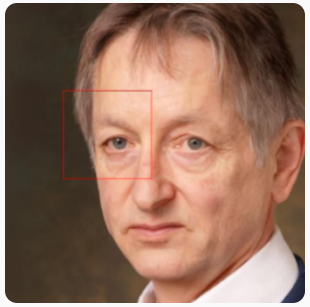
The same loop runs today's image generators



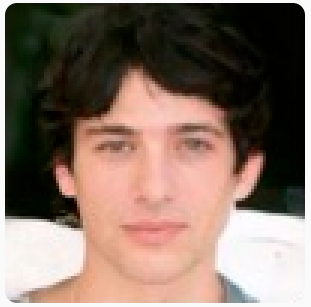
"a basket of cherries"



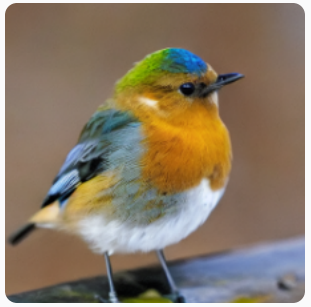
fill a hole



sharpen a blurry face



a sketch → a scene



edges → a bird



your dog, at the Acropolis

published Video, audio, 3-D shapes, protein structures and robot motions are made the same way.

> The recipe goes beyond images: video, audio, proteins, robot policies. By the end you can say what each network was trained to predict.

Published samples: LDM (Rombach et al. 2022) · RePaint (Lugmayr et al. 2022) · SR3 (Saharia et al. 2022) · SDEdit (Meng et al. 2022) · ControlNet (Zhang et al. 2023) · DreamBooth (Ruiz et al. 2023). Showcase after MIT 6.S183 "A Practical Introduction to Diffusion Models" (IAP 2025), Lecture 1

▶ Notebook 0.2

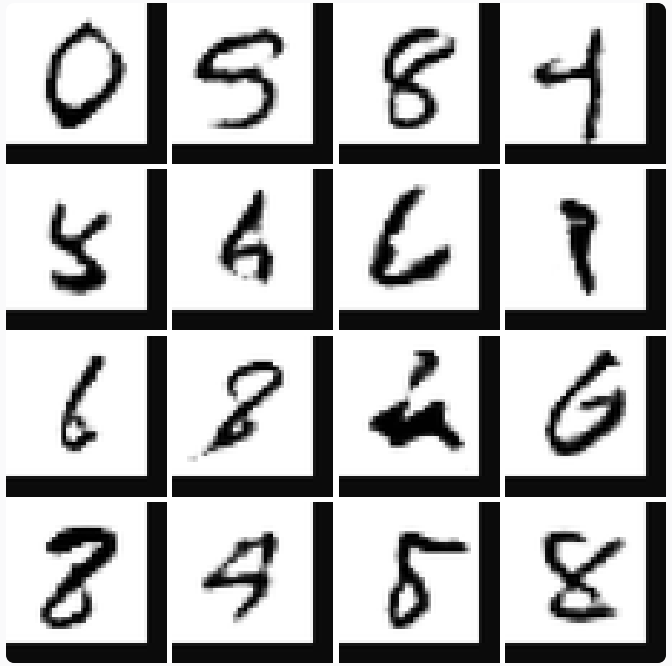
Two generators so far, and a third idea

VAE (Lecture 3)



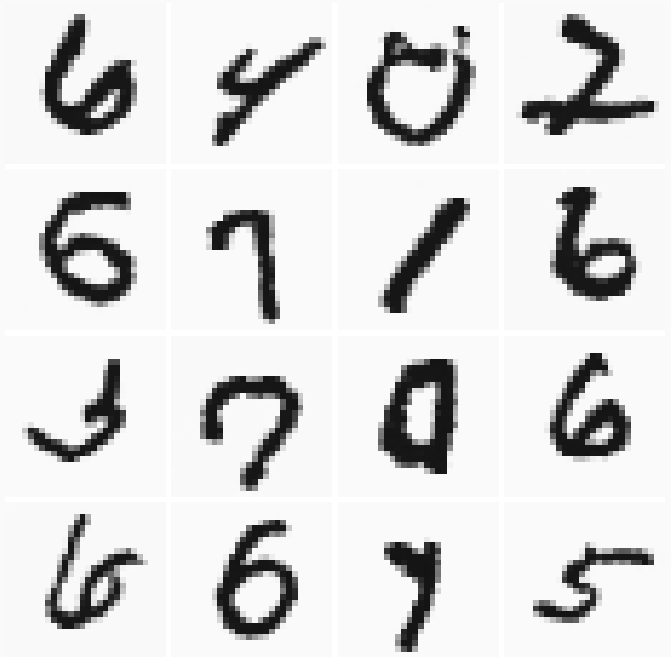
one decoder step from z: honest likelihood, blurry digits

GAN (Lecture 4)



one generator step: sharp, but a fragile game that can collapse

Diffusion (today)



a thousand small denoising steps, each one an easy regression

> Diffusion makes a digit in a thousand small steps. Each step is a simple regression problem, and the result is sharp.

▶ Notebook 0.3

The idea in one sentence

If you learn to undo each tiny step of a destruction process, you can create from scratch.

Paras Chopra, “Diffusion Models from First Principles”



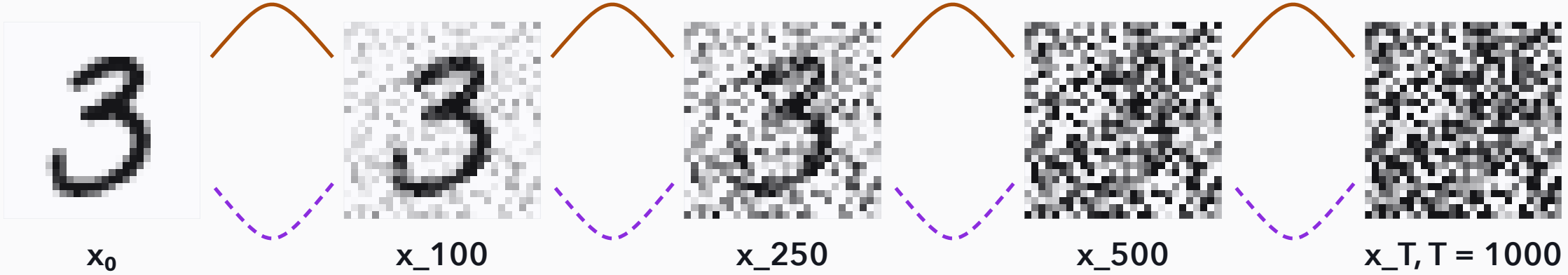
illustration Generated with Gemini. A drop of ink spreads until the water is uniform; nothing pulls it back together.

> Film a drop of ink spreading through water. Played forwards, structure disappears. Between two neighbouring frames the change is tiny, and tiny changes can be predicted. Learn to undo each one, and you can run the film backwards.

▶ Notebook 0.3

The plan: destroy, step back, train, generate

q: add noise, step by step. Fixed: nothing to learn.



p_θ: remove noise, step by step. Learned: today's network.

> Right to left: remove a little noise at a time. That direction is learned (Parts 2–4). Then: steering with labels and text (5), faster sampling (6), and what the network really learns (7).

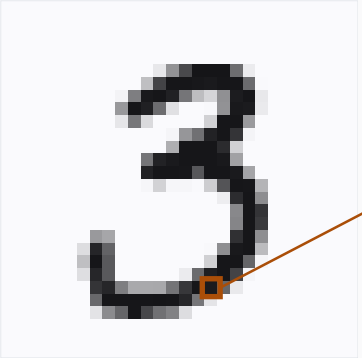
Ho, Jain & Abbeel, "Denoising Diffusion Probabilistic Models" (2020)

▶ Notebook 1.1

One noising step: shrink a little, add a little noise

$$x_t = \sqrt{\alpha_t}\, x_{t-1} + \sqrt{\beta_t}\, \varepsilon_t$$

$$\alpha_t = 1 - \beta_t, \qquad \varepsilon_t \sim \mathcal{N}(0,1)$$



pixel (22, 16)

its value x_0 = +1: ink

paper is −1, ink is +1



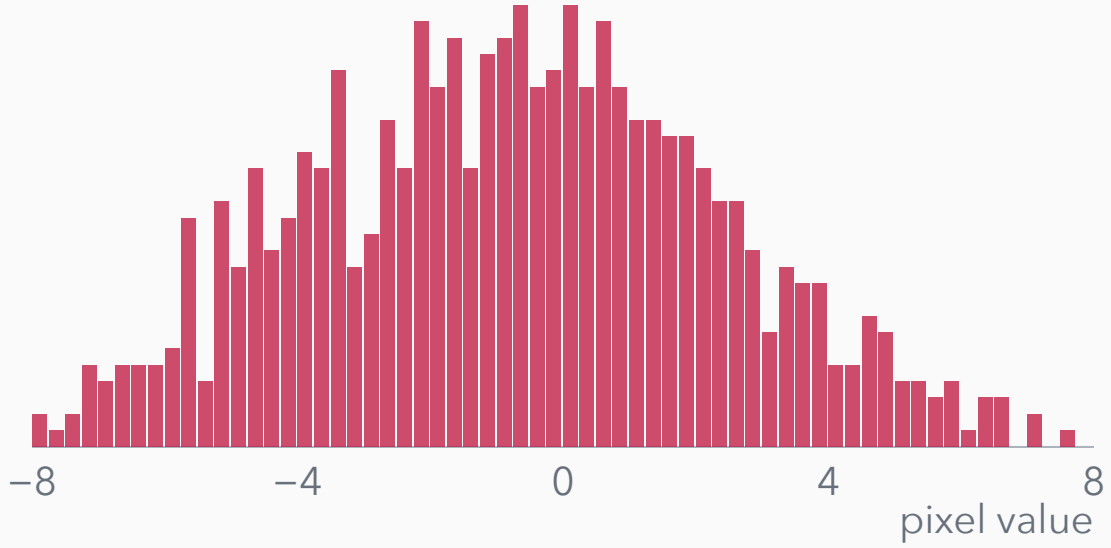
β_t is how much noise step t adds. Real models take 1,000 tiny steps; to see every number we first use a toy with five big steps.

> Repeat with a growing β. But first: why shrink at all?

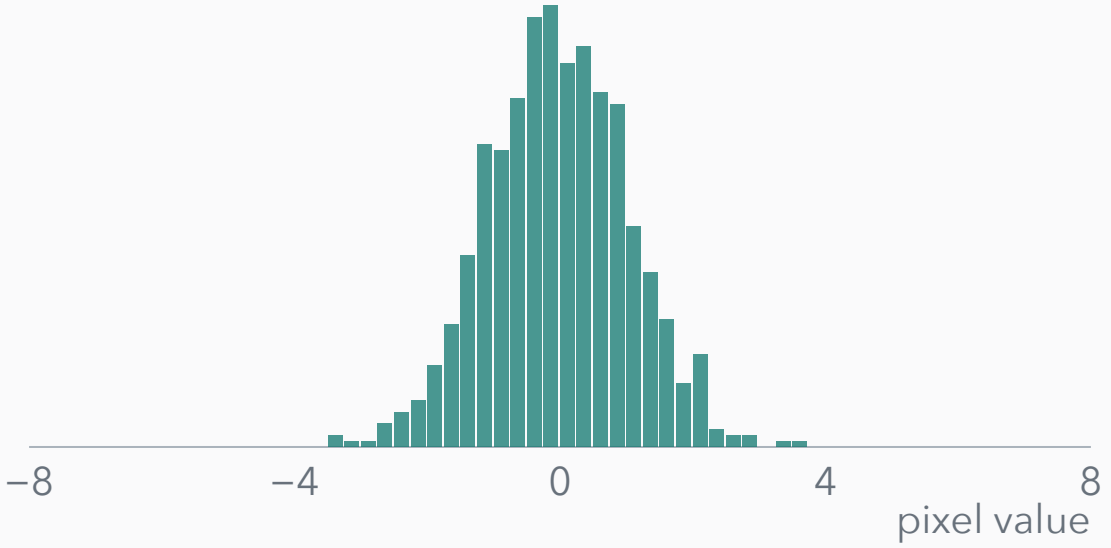
▶ Notebook 1.1

Keep adding noise: what happens to the spread?

add noise only spread (sd) = 3.22



shrink a little, then add spread (sd) = 1.05



> Now shrink a little before adding: multiply by $\sqrt{(1-\beta)}$, then add $\sqrt{\beta}\,\varepsilon$. The spread settles at 1 and stays there.

Deepia, "Diffusion Models: DDPM, Generative AI Animated" (2025)

▶ Notebook 1.1

Shrink a little, then add: the spread stays at 1

$$x_t = \sqrt{1-\beta_t}\, x_{t-1} + \sqrt{\beta_t}\, \varepsilon_t$$

$$\mathrm{Var}(x_t) = (1-\beta_t)\,\mathrm{Var}(x_{t-1}) + \beta_t$$

Rule: Var(aX + bε) = a² Var X + b² Var ε when X and ε are independent, and Var ε = 1.

$$\mathrm{Var}(x_{t-1}) = 1 \;\Rightarrow\; \mathrm{Var}(x_t) = (1-\beta_t) + \beta_t = 1$$

And if the variance is not yet 1, the gap shrinks: $\mathrm{Var}(x_t) - 1 = (1-\beta_t)(\mathrm{Var}(x_{t-1}) - 1)$. Our pixel's first step: $\sqrt{0.9999} \times 1 + 0.01 \times (-0.5) = 0.995$.

> If the variance was 1, it stays exactly 1. If it was not, every step moves it closer to 1.

▶ Notebook 1.2

A toy schedule you can do by hand: five steps

t	β_t	$\alpha_t = 1 - \beta_t$	$\bar{\alpha}_t = \bar{\alpha}_{t-1} \times \alpha_t$	signal $\sqrt{\bar{\alpha}_t}$	noise $\sqrt{(1 - \bar{\alpha}_t)}$
1	0.1	$1 - 0.1 = 0.9$	$\alpha_1 =$ **0.9**	0.949	0.316
2	0.2	$1 - 0.2 = 0.8$	$0.9 \times 0.8 =$ **0.72**	0.849	0.529
3	0.3	$1 - 0.3 = 0.7$	$0.72 \times 0.7 =$ **0.504**	0.710	0.704
4	0.5	$1 - 0.5 = 0.5$	$0.504 \times 0.5 =$ **0.252**	0.502	0.865
5	0.8	$1 - 0.8 = 0.2$	$0.252 \times 0.2 =$ **0.0504**	0.224	0.974

> The two numbers we use all lecture: $\sqrt{\bar{\alpha}_t}$, the size of the signal, and $\sqrt{(1 - \bar{\alpha}_t)}$, the size of the noise. By step 5 they are 0.224 and 0.974: almost pure noise.

▶ Notebook 1.2

Forward, step by step: every number

t	x_{t-1}	ε_t	$\sqrt{a_t} \cdot x_{t-1} + \sqrt{\beta_t} \cdot \varepsilon_t$	x_t
1	1.000	0.3	0.949 × 1.000 + 0.316 × 0.3	**1.044**
2	1.044	−0.5	0.894 × 1.044 + 0.447 × (−0.5)	**0.710**
3	0.710	0.6	0.837 × 0.710 + 0.548 × 0.6	**0.922**
4	0.922	−1.2	0.707 × 0.922 + 0.707 × (−1.2)	**−0.196**
5	−0.196	0.5	0.447 × (−0.196) + 0.894 × 0.5	**0.359**

From $x_5 = 0.359$ alone you cannot tell the pixel was ink. (A paper pixel with the same draws would give −0.090.)

> Step 5: $x_5 = 0.359$. The original value is almost gone.

▶ Notebook 1.3

Do we need a 1,000-step loop for one noisy example?

$$x_1 = \sqrt{\alpha_1}\, x_0 + \sqrt{\beta_1}\, \varepsilon_1 \qquad (\alpha_t = 1 - \beta_t)$$

$$x_2 = \sqrt{\alpha_2}\left(\sqrt{\alpha_1}\, x_0 + \sqrt{\beta_1}\, \varepsilon_1\right) + \sqrt{\beta_2}\, \varepsilon_2$$

$$x_3 = \sqrt{\alpha_3}\left(\sqrt{\alpha_2}\left(\sqrt{\alpha_1}\, x_0 + \sqrt{\beta_1}\, \varepsilon_1\right) + \sqrt{\beta_2}\, \varepsilon_2\right) + \sqrt{\beta_3}\, \varepsilon_3 \cdots$$

> Three steps, and it keeps nesting. Training needs millions of noisy examples at random t: we cannot afford this loop each time.

Ari Seff, "What are Diffusion Models?" (2022)

▶ Notebook 1.3

Two steps by hand: two noises become one

$$x_2 = \underbrace{\sqrt{\alpha_2\alpha_1}\,x_0}_{\text{signal}} + \underbrace{\sqrt{\alpha_2\beta_1}\,\varepsilon_1 + \sqrt{\beta_2}\,\varepsilon_2}_{\text{two independent Gaussians}}$$

$$\mathrm{Var}(\text{noise}) = \alpha_2\beta_1 + \beta_2 = 1 - \alpha_1\alpha_2$$

$$x_2 = \sqrt{\alpha_1\alpha_2}\,x_0 + \sqrt{1-\alpha_1\alpha_2}\,\varepsilon, \qquad \varepsilon \sim \mathcal{N}(0,1)$$

$$\beta_1 = 0.1,\ \beta_2 = 0.2: \quad x_2 = \sqrt{0.72}\,(1) + \sqrt{0.28}\,(-0.26) = 0.71$$

> Check with two big illustrative steps, $\beta_1 = 0.1$ and $\beta_2 = 0.2$, and the draws $\varepsilon_1 = 0.3$, $\varepsilon_2 = -0.5$: step by step $x_2 = 0.71$, and the formula with the combined $\varepsilon = -0.26$ gives the same.

Ari Seff, "What are Diffusion Models?" (2022)

▶ Notebook 1.3

The shortcut: jump straight to any step

$$x_t = \sqrt{\bar{\alpha}_t}\, x_0 + \sqrt{1-\bar{\alpha}_t}\, \varepsilon, \qquad \bar{\alpha}_t = \alpha_1 \alpha_2 \cdots \alpha_t$$

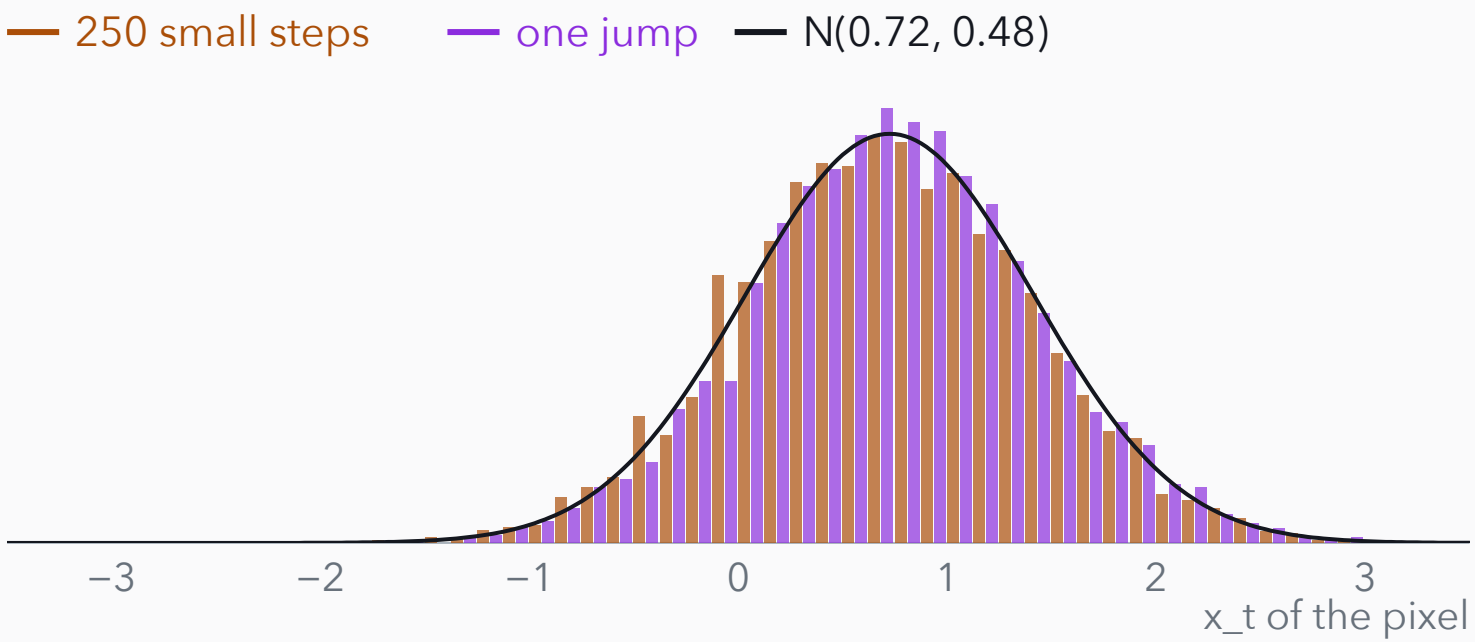
$$\text{toy}, t = 2: \quad \sqrt{0.72} \times 1 + \sqrt{0.28} \times (-0.262) = 0.710 \qquad (\text{the same as two steps})$$

```
def q_sample(x0, t, eps, alpha_bar):
    a = alpha_bar[t]                              # one number per example
    return a.sqrt() * x0 + (1 - a).sqrt() * eps
```

> In code: one line. Any noise level, any image, in one shot.

▶ Notebook 1.3

Step by step or one jump: the same distribution



t = 250, 4,000 runs each

steps: mean 0.713, sd 0.699

jump: mean 0.735, sd 0.679

exact: mean 0.724, sd 0.690

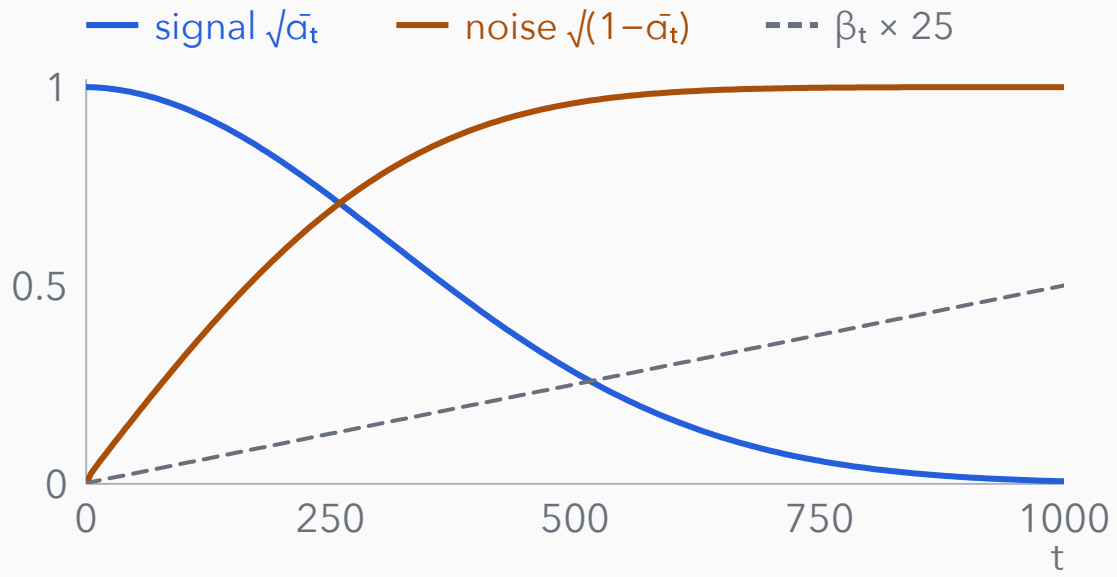
250 noise draws per run, or one

> 4,000 runs each way, from our ink pixel: t small steps with their own noise draws, or one jump with one draw. Same histogram, same mean and spread, matching the formula. Move t.

Paras Chopra, "Diffusion Models from First Principles" (2025)

▶ Notebook 1.4

The real schedule: 1,000 small steps



$$t = 250: \quad \bar{\alpha} = 0.524, \quad x_{250} = 0.724 \times 1 + 0.690 \times (-0.5) = 0.379$$

> Our running example from now on: t = 250, where signal and noise are about equal. The ink pixel with the noise draw ε = −0.5 becomes x_{250} = 0.379.

Ho, Jain & Abbeel, "Denoising Diffusion Probabilistic Models" (2020) · Nichol & Dhariwal, "Improved Denoising Diffusion Probabilistic Models" (2021)

▶ Notebook 1.4

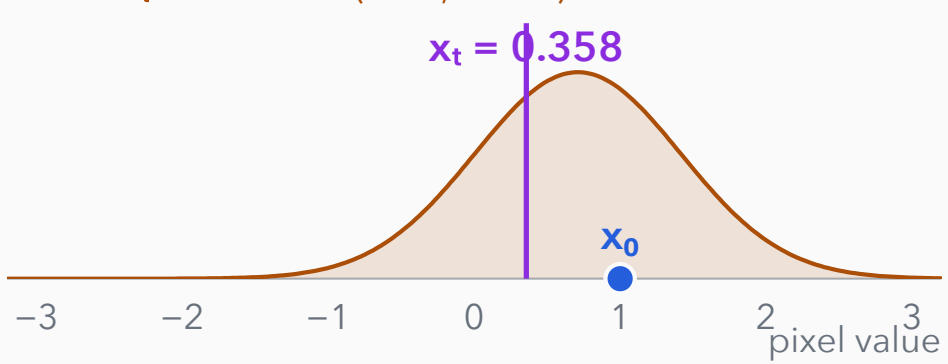
Try it: every number of a forward jump

$$\beta_3 = 0.3, \qquad \alpha_3 = 1 - \beta_3 = 0.7$$

$$\bar{\alpha}_3 = \alpha_1 \alpha_2 \alpha_3 = 0.9 \times 0.8 \times 0.7 = 0.5040$$

$$\sqrt{\bar{\alpha}_3} = 0.710, \qquad \sqrt{1 - \bar{\alpha}_3} = 0.704$$

$$x_3 = \sqrt{\bar{\alpha}_3}\, x_0 + \sqrt{1 - \bar{\alpha}_3}\, \varepsilon = 0.710 \times 1.000 + 0.704 \times (-0.500) = \mathbf{0.358}$$

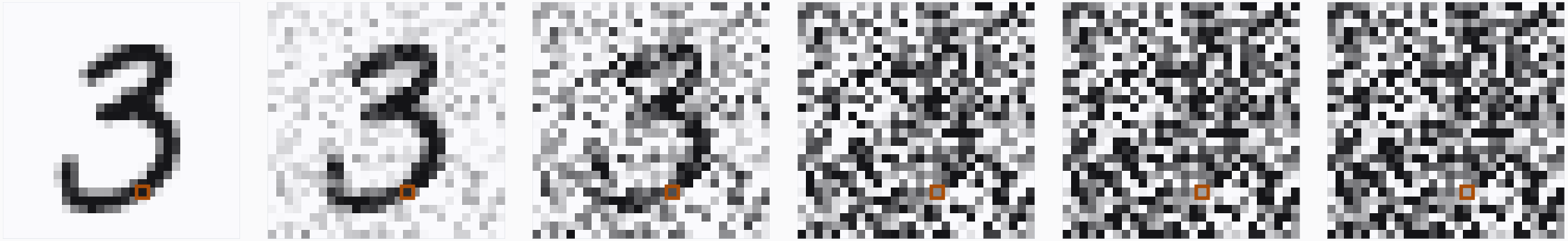


t	β_t	$\alpha_t = 1 - \beta_t$	$\bar{\alpha}_t$	$\sqrt{\bar{\alpha}_t}$	$\sqrt{1 - \bar{\alpha}_t}$
1	0.1	0.9	0.9000	0.949	0.316
2	0.2	0.8	0.7200	0.849	0.529
3	**0.3**	**0.7**	**0.5040**	**0.710**	**0.704**
4	0.5	0.5	0.2520	0.502	0.865
5	0.8	0.2	0.0504	0.224	0.974

Pick the toy or the real schedule, a step t, ink or paper, and a noise draw ε. Every number is shown: β, α, the product ᾱ, the two coefficients and the noisy value, with the bell of all values x_t could take.

▶ Notebook 1.5

Slide t: watch our 3 dissolve



t = 0	t = 100	t = 250	t = 500	t = 750	t = 1000
$\sqrt{\bar{\alpha}} = 1.00$	$\sqrt{\bar{\alpha}} = 0.95$	$\sqrt{\bar{\alpha}} = 0.72$	$\sqrt{\bar{\alpha}} = 0.28$	$\sqrt{\bar{\alpha}} = 0.06$	$\sqrt{\bar{\alpha}} = 0.01$

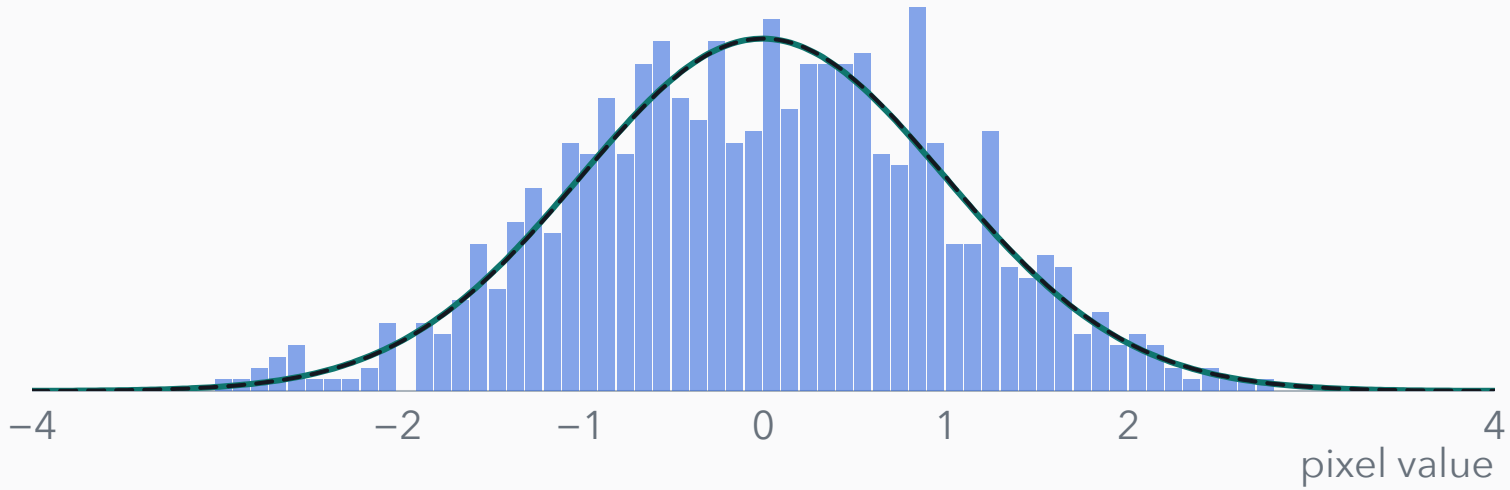
> Around t = 600 the 3 is gone: $\sqrt{\bar{\alpha}}$ has fallen to 0.16. Our ink pixel (+1) reads 0.00 at t = 500: from the value alone you can no longer tell ink from paper.

Paras Chopra, "Diffusion Models from First Principles" (2025)

▶ Notebook 1.5

Every image ends in the same place: N(0, 1)

— our 3: its 784 pixels — all MNIST pixels (exact) --- N(0, 1)



t = 1000

signal $\sqrt{\bar{\alpha}} = 0.006$

noise $\sqrt{(1 - \bar{\alpha})} = 1.000$

only the noise term is left

> Why a Gaussian? Not the Central Limit Theorem: the signal part $\sqrt{\bar{\alpha}_t}\, x_0$ is shrunk to zero, and what is left, $\sqrt{(1 - \bar{\alpha}_t)}\, \varepsilon$, is Gaussian because we made it Gaussian.

Paras Chopra, "Diffusion Models from First Principles" (2025) · (endpoint argument corrected)

▶ Notebook 1.5

Destroying is easy, and it hands us labels for free

we have	how
a clean image x_0	the dataset
a noise level t	pick 1 … 1,000 at random
the noise ε	draw N(0, I)
the noisy image x_t	one line: $\sqrt{\bar{\alpha}_t}\, x_0 + \sqrt{(1 - \bar{\alpha}_t)}\, \varepsilon$

Next: can we go back? Given x_t, where was x_{t-1}?

> The hard direction is the other one. Part 2: undoing it, one step back.

▶ Notebook 2.1

Going back: where was x_{t-1}?

$$q(x_{t-1} \mid x_t) = \frac{q(x_t \mid x_{t-1})\, q(x_{t-1})}{q(x_t)}$$

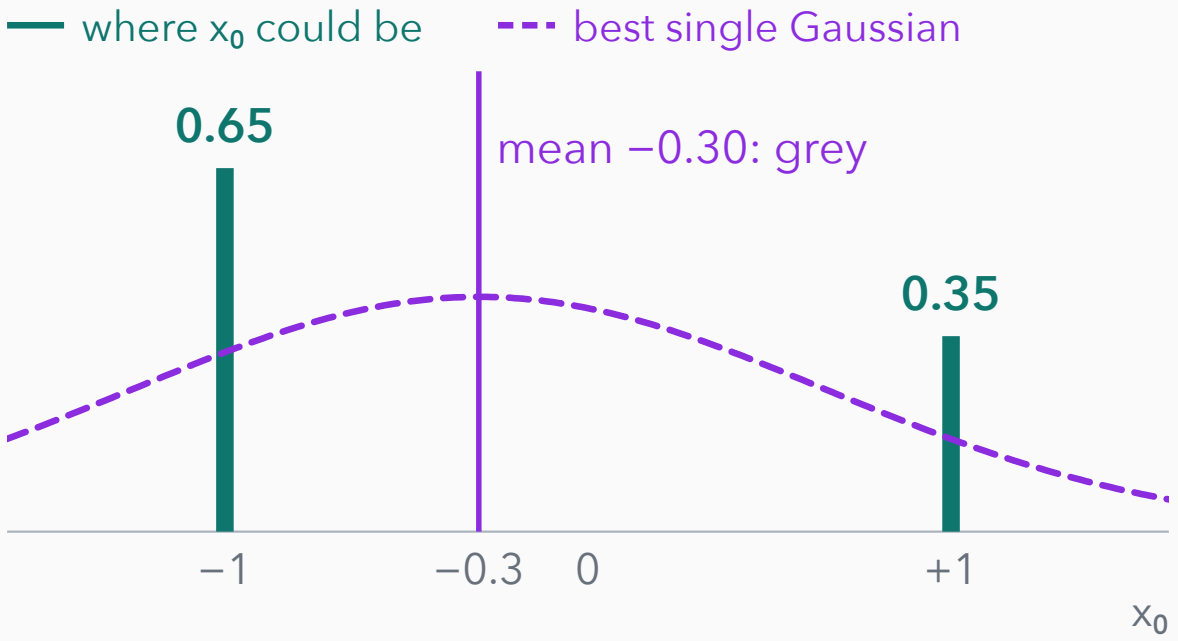
term	what it is	known?
$q(x_t \mid x_{t-1})$	our noising step: $N(\sqrt{\alpha_t}\, x_{t-1}, \beta_t)$	yes
$q(x_{t-1})$	the distribution of all noisy images at step t − 1	no

For images we cannot write that prior down. For one pixel, we can.

> So we cannot compute the step back exactly for images. For one pixel we can, and it teaches us what to learn.

▶ Notebook 2.2

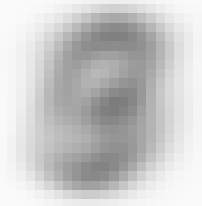
From pure noise straight back to the pixel: two spikes



One Gaussian has one bump: its best fit sits at the average, a grey that is neither paper nor ink.



average of 6,131 training 3s



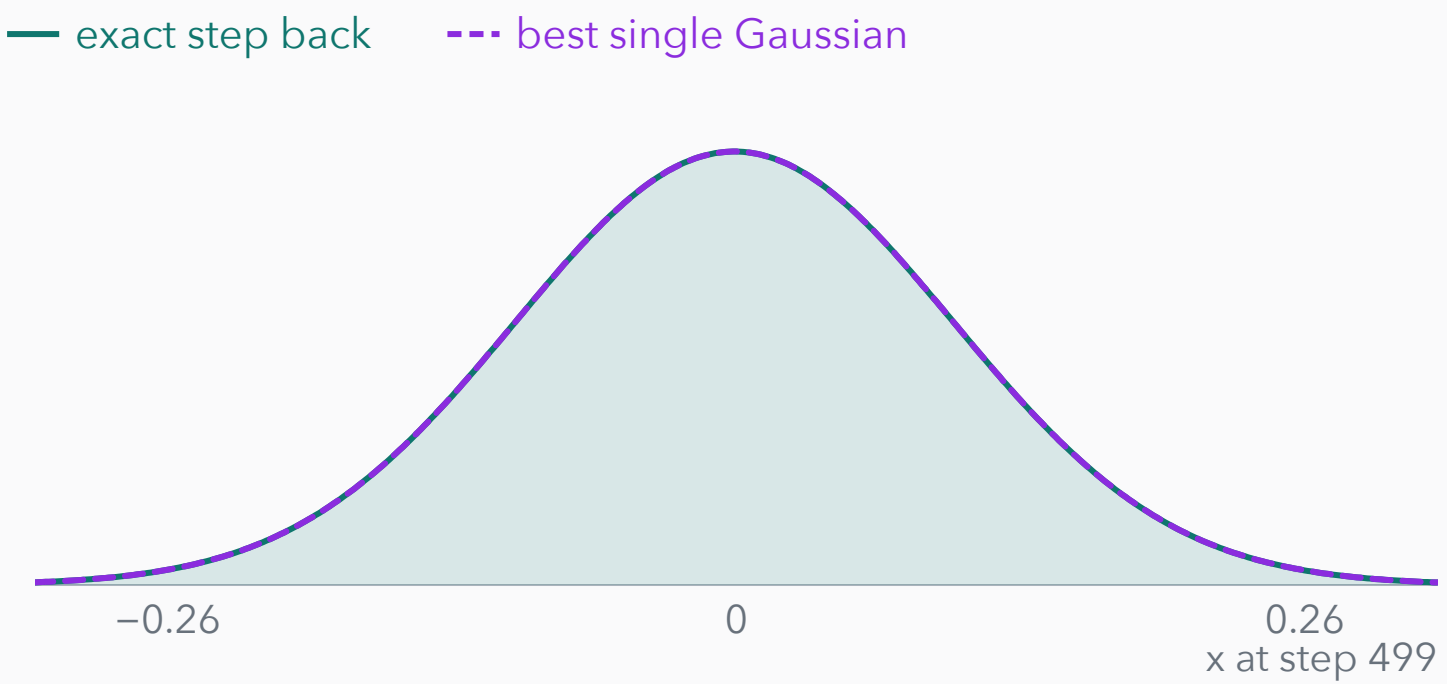
average of all digits

> For whole images the same thing happens: a one-step denoiser asked for “a digit” returns the average digit, a blur. Average a hundred cats and you get no cat at all.

MIT 6.S183 “A Practical Introduction to Diffusion Models” (IAP 2025), Lecture 2 (the “100 cats” question)

▶ Notebook 2.2

Shrink the jump: the step back becomes one Gaussian



from t = 500 back to 499

a jump of 1 step

one bell: a Gaussian fits

bump centres 0.1 widths apart

observed x_t = 0.00

> One step back, 500 → 499: the bumps are 0.06 widths apart, one bell. Press play to shrink the jump continuously.

Ari Seff, "What are Diffusion Models?" (2022) · (the 1-D picture, here computed exactly)

▶ Notebook 2.3

If we knew the noise, we could undo the jump exactly

$$x_t = \sqrt{\bar{\alpha}_t}\, x_0 + \sqrt{1-\bar{\alpha}_t}\, \varepsilon \quad \Longleftrightarrow \quad x_0 = \frac{x_t - \sqrt{1-\bar{\alpha}_t}\, \varepsilon}{\sqrt{\bar{\alpha}_t}}$$

$$\text{toy}, t = 2: \quad \frac{0.710 - \sqrt{0.28} \times (-0.262)}{\sqrt{0.72}} = 1.000$$

$$\hat{\varepsilon} = \text{network}(x_t,\, t) \;\approx\; \varepsilon$$

> When generating, nobody tells us ε. So a network guesses it from what it can see: x_t and t. Everything else is arithmetic.

▶ Notebook 2.3

Three ways to say the same guess

the network outputs	meaning	our pixel (best guess)
$\hat{\varepsilon}$	the noise: "what is the noise? subtract it"	0.28
$\hat{x}_0$	the clean value: "what is under the noise?"	0.26
$s = \nabla_x \log q_t$	the score: "which way is more likely?" (Part 7)	−0.40

Each can be converted into the others with the shortcut. DDPM predicts ε.

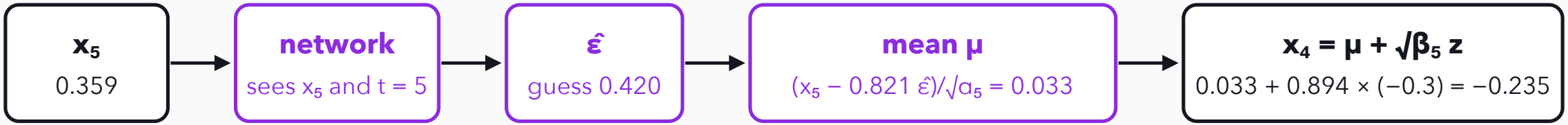
> So the network has one job: guess the noise. With that guess, one step back is simple arithmetic: next slide.

Paras Chopra, "Diffusion Models from First Principles" (2025) · Outlier, "Diffusion Models, Paper Explanation, Math Explained" (2022) (4:48)

▶ Notebook 2.4

One step back, in one line

$$x_{t-1} = \underbrace{\frac{1}{\sqrt{\alpha_t}}\left(x_t - \frac{\beta_t}{\sqrt{1-\bar{\alpha}_t}}\,\hat{\varepsilon}\right)}_{\mu\text{: a little less noise}} + \underbrace{\sqrt{\beta_t}\,z}_{\text{fresh noise}}, \qquad z \sim \mathcal{N}(0,1)$$



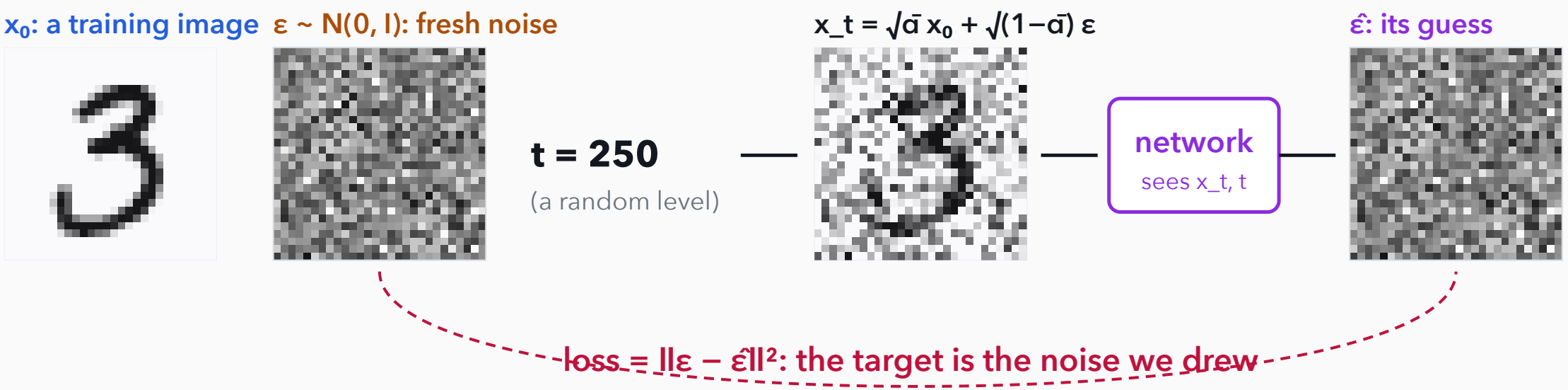
Only the purple box is learned. The fresh noise z is drawn, like ε in the forward step.

> That is the whole generator: start from pure noise and repeat this step from t = T down to 1. Only the network is learned; everything else is the schedule's numbers.

Ho, Jain & Abbeel, "Denoising Diffusion Probabilistic Models" (2020) (Algorithm 2) · Paras Chopra, "Diffusion Models from First Principles" (2025) (V)

▶ Notebook 3.1

A training example costs one line



> The network sees x_t and t and guesses the noise. The answer is the ε we drew ourselves: labels for free, as many as we like.

Ho, Jain & Abbeel, "Denoising Diffusion Probabilistic Models" (2020) (Algorithm 1)

▶ Notebook 3.1

ε is all the noise since x_0, not the last step's

$$x_t = \sqrt{\bar{\alpha}_t}\, x_0 + \underbrace{\sqrt{1-\bar{\alpha}_t}\, \varepsilon}_{\text{every step's noise, combined}}$$

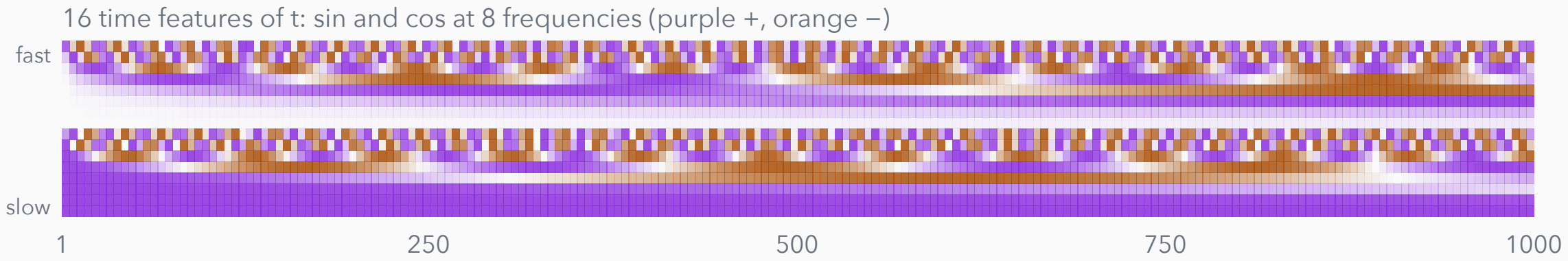
t = 250	size
all noise added since x_0: $\sqrt{(1-\bar{\alpha})}$	0.690
the last step's noise alone: $\sqrt{\beta_{250}}$	0.071

> At t = 250 the last step's noise has size 0.071; all the noise together, 0.690. The network predicts the whole thing.

▶ Notebook 3.2

One network for every noise level: tell it t

the same input $x_t = 0.38$	best guess of x_0
at t = 50	1.00: ink, surely
at t = 250	0.26: unsure
at t = 900	−0.29: just the prior



> Like token positions in a transformer, t becomes a vector of sines and cosines at different frequencies, so nearby steps look alike and far ones differ.

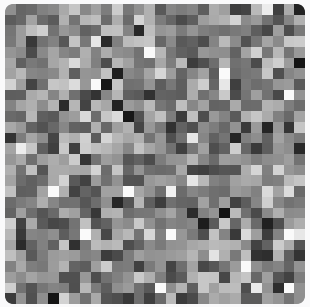
Outlier, "Diffusion Models, Paper Explanation, Math Explained" (2022) (7:01) · MIT 6.S183 "A Practical Introduction to Diffusion Models" (IAP 2025), Lecture 1 (22:21)

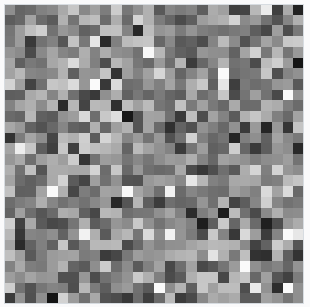
▶ Notebook 3.3

The loss: squared error between the noise and its guess

$$L(\theta) = \left\| \varepsilon - \hat{\varepsilon}_\theta(x_t, t) \right\|^2 \quad \text{(averaged over pixels and examples)}$$

$$\text{pixel:} \quad (-0.5 - 0.28)^2 = 0.60$$





ε: the noise we drew ε̂: the network's guess, t = 250 mean squared error 0.024

> Our 3 at t = 250 through the trained network: the guessed noise looks just like the real one; the average squared error over 784 pixels is 0.024.

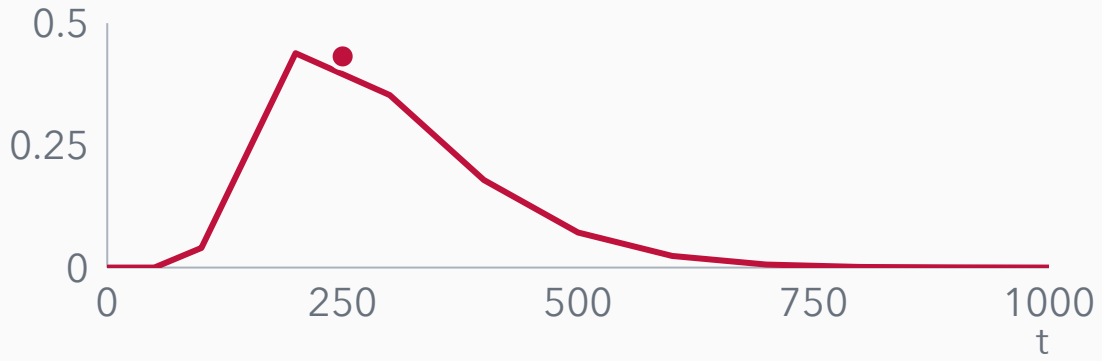
▶ Notebook 3.3

The same noisy value can come from different noises

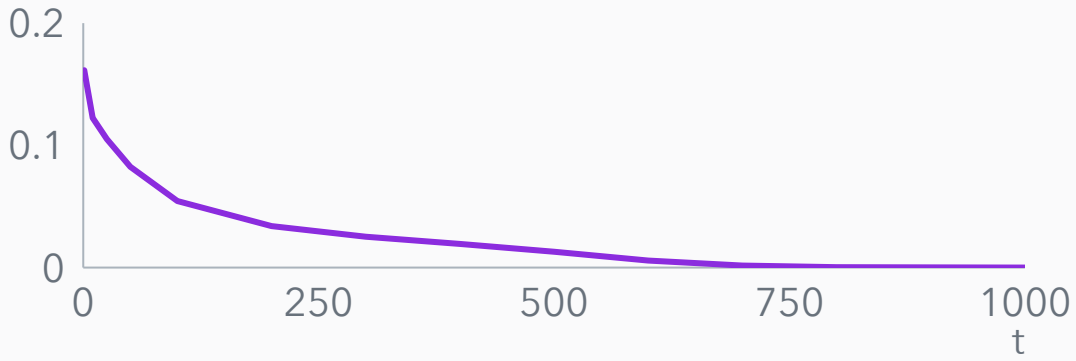
$$x_{250} = 0.38: \quad \underbrace{\text{ink},\ \varepsilon = -0.50}_{P=0.63} \quad \text{or} \quad \underbrace{\text{paper},\ \varepsilon = 1.60}_{P=0.37}$$

$$\hat{\varepsilon}^* = \mathbb{E}[\varepsilon \mid x_t] = 0.63 \times (-0.50) + 0.37 \times 1.60 = 0.28$$

one pixel: the least possible loss



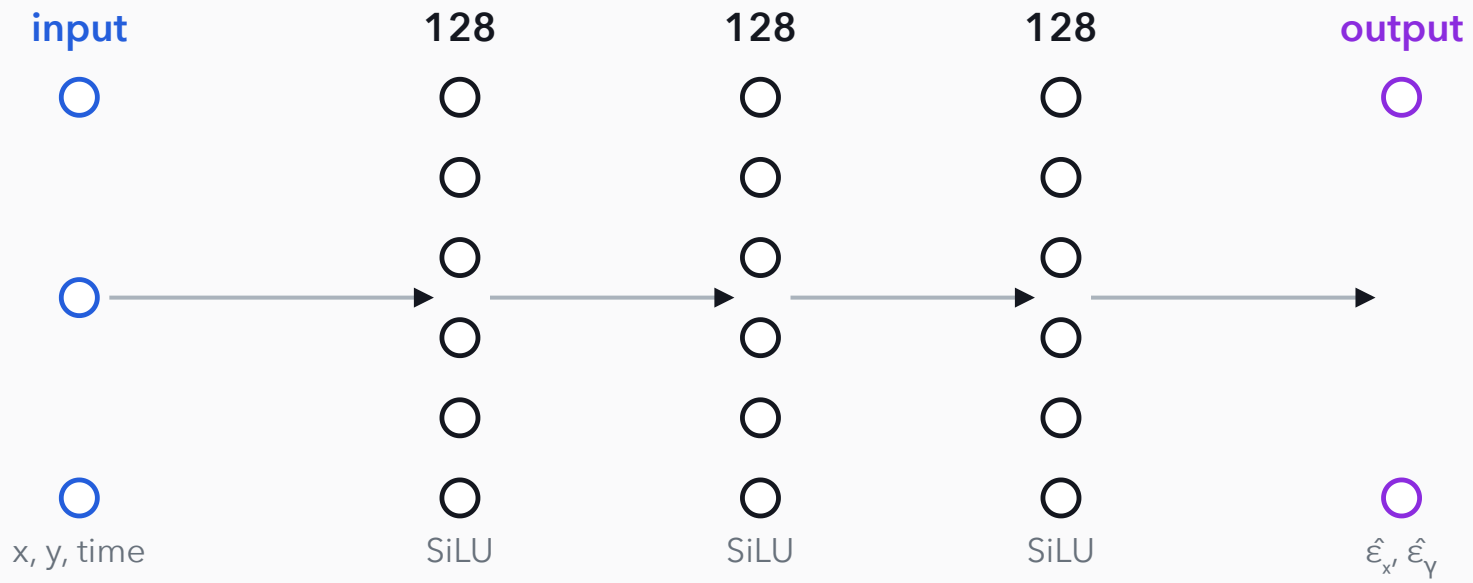
our MNIST network: its loss at each t



So the loss cannot reach zero. Left: for our pixel the unavoidable loss is largest in the middle, where both stories are plausible. Right: our trained network's loss by noise level.

▶ Notebook 3.4

For points in 2-D: a small MLP



the data: 2-D points



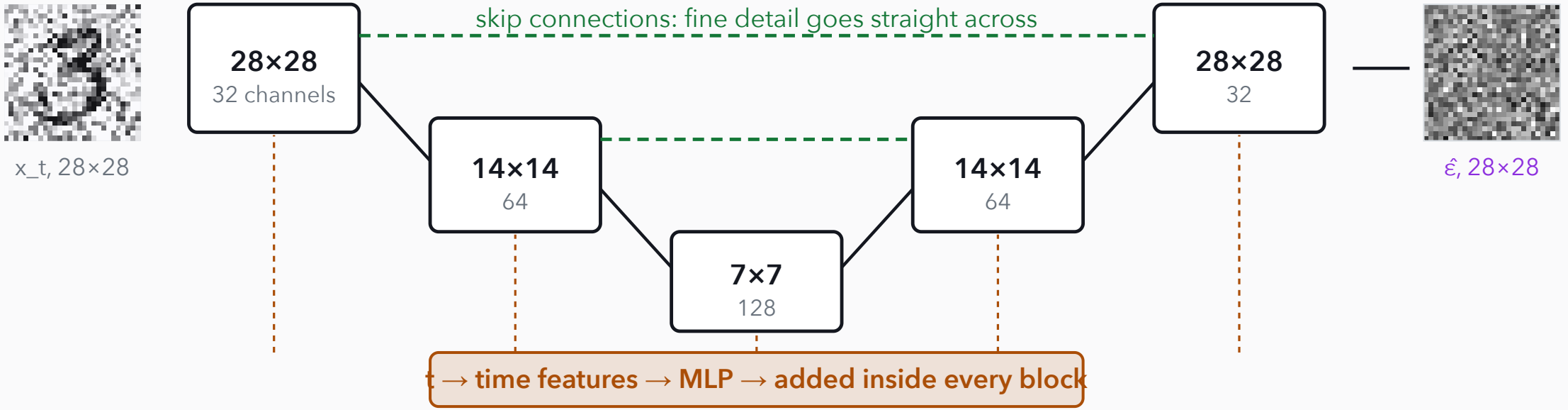
Our 2-D example: points on a smiley face. 35,714 parameters; the time features are simply concatenated to the input.

> Before images, a dataset we can see completely: points on a smiley face. The network: the point (x, y) and the 16 time features in, three layers of 128, two numbers out: the guessed noise.

▶ Notebook 3.4

For images: a U-Net, so every pixel can see the whole digit

1 · Destroy | 2 · Undo | **3 · Train** | 4 · Generate | 5 · Steer | 6 · Faster | 7 · What it learns



> The time embedding is added inside every block: the same weights, told how noisy the input is.

Ho, Jain & Abbeel, "Denoising Diffusion Probabilistic Models" (2020) · Outlier, "Diffusion Models, Paper Explanation, Math Explained" (2022) (7:01)

▶ Notebook 3.5

The training step in eight lines


```
t   = torch.randint(1, T + 1, (B,))          # a level per image
eps = torch.randn_like(x0)                   # the noise: also the target
a   = alpha_bar[t].view(B, 1, 1, 1)
xt  = a.sqrt() * x0 + (1 - a).sqrt() * eps   # the shortcut
eps_hat = model(xt, t)                       # the guess
loss = ((eps - eps_hat) ** 2).mean()         # squared error
opt.zero_grad(); loss.backward()
opt.step()                                   # one gradient step
```

That is all of training: no adversary, no encoder, no sampling. Our MNIST model ran 30,000 such steps (batch 128), about 50 minutes on a laptop GPU.

> Compare with the GAN of Lecture 4: one network, one loss, no game. Diffusion pays later, at sampling time.

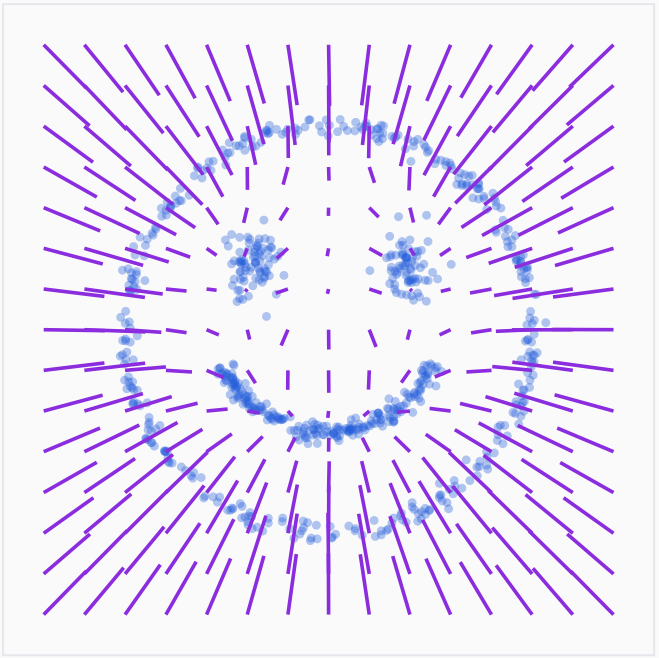
Ho, Jain & Abbeel, "Denoising Diffusion Probabilistic Models" (2020) (Algorithm 1)

▶ Notebook 3.5

Press play: the network learns the smiley

1 · Destroy | 2 · Undo | **3 · Train** | 4 · Generate | 5 · Steer | 6 · Faster | 7 · What it learns

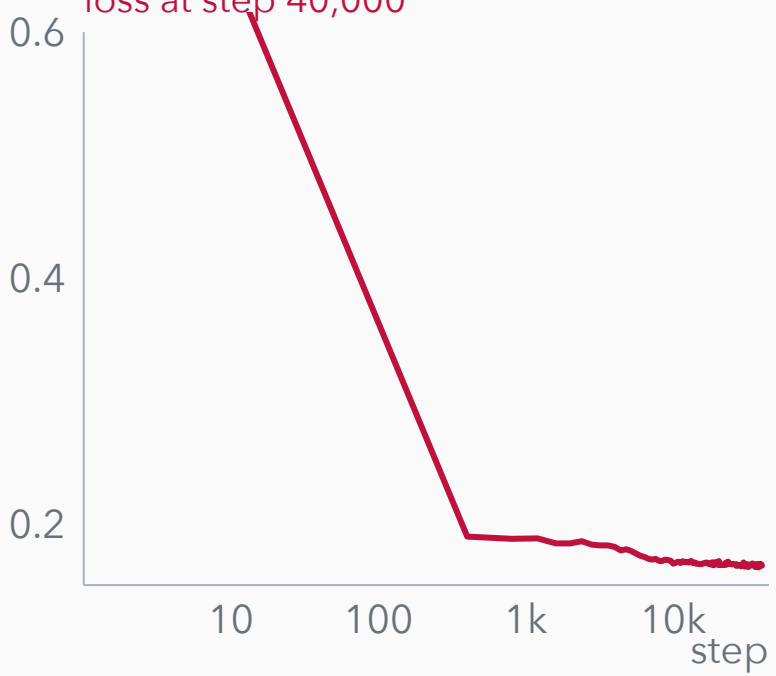
arrows: $x_t \rightarrow$ the network's $\hat{x}_0$ (t = 100)



300 samples from this network



loss at step 40,000

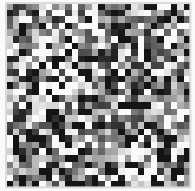


> Press play: as the loss falls the arrows turn towards the face, and the samples, drawn with Part 4's loop, become a smiley.

Paras Chopra, "Diffusion Models from First Principles" (2025) ("Training Playground", here with a real network)

▶ Notebook 3.6

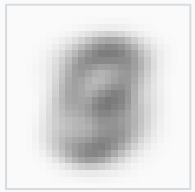
What the network believes our 3 is, at each noise level



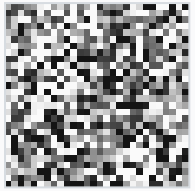
t = 1000



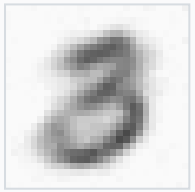
$\hat{x}_0$ error 0.756



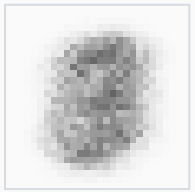
alone 0.248



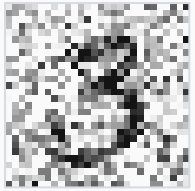
t = 600



$\hat{x}_0$ error 0.154



alone 0.240



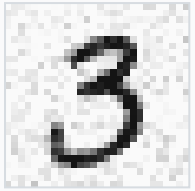
t = 250



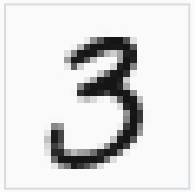
$\hat{x}_0$ error 0.021



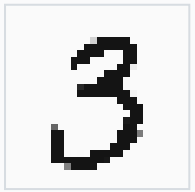
alone 0.131



t = 50



$\hat{x}_0$ error 0.003



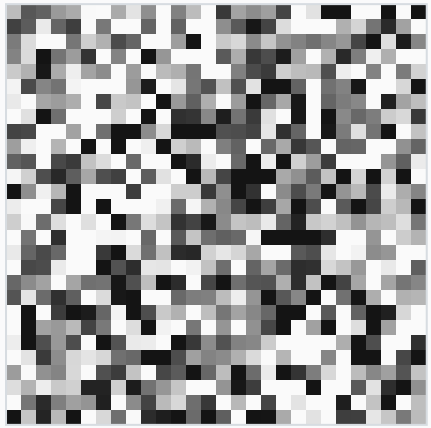
alone 0.041

> Each pixel guessed on its own (the best one-pixel guess, from optional depth) stays a grey blur much longer. The network's guesses are far better.

▶ Notebook 3.6

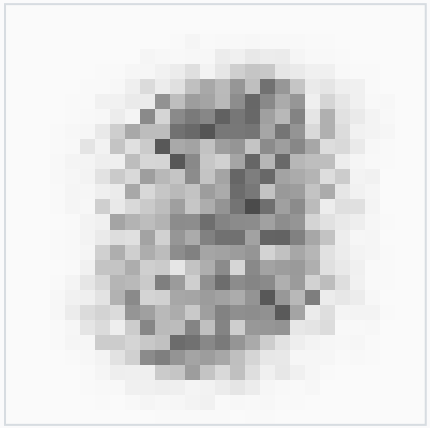
One pixel alone gives a grey; all 784 together give a 3

x_{500}



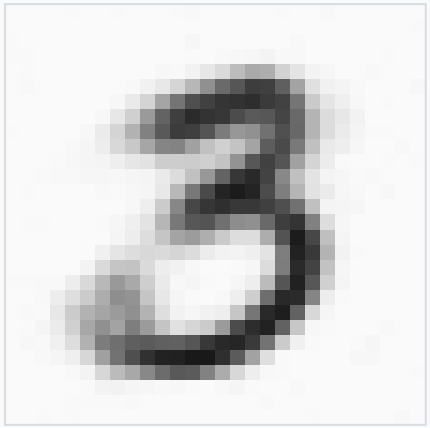
the input

each pixel alone



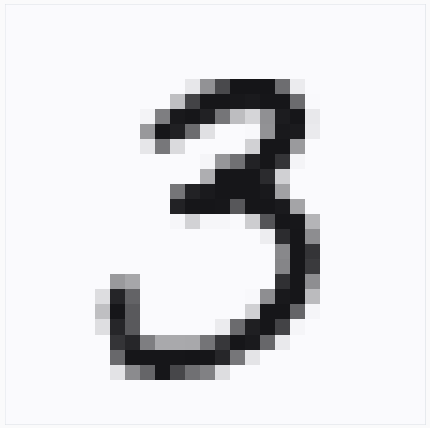
error 0.230

the network



error 0.075

our 3



> The network looks at all of them at once and returns a clean 3: error 0.075, three times smaller. Lecture 2's lesson again: strokes are coordinated choices, and the U-Net can see them.

▶ Notebook 3.6

Training: a regression problem we made ourselves

step	what happens
draw	an image x_0, a level t, a noise ε
make	x_t with the shortcut
guess	$\hat{\varepsilon} = \text{network}(x_t, t)$
learn	minimise $\|\varepsilon - \hat{\varepsilon}\|^2$: the network learns $E[\varepsilon \mid x_t]$, the best average guess

Next: generate. Start from pure noise and use the guesses to walk back, one small Gaussian step at a time.

> Part 4: put the guess into Part 2's step back, and repeat a thousand times.

▶ Notebook 4.1

One step back, by hand, on the real schedule

$$\hat{\varepsilon} = \text{network}(x_{250} = 0.379,\ t = 250) = 0.276$$

$$\mu = \frac{1}{\sqrt{0.9949}}\left(0.379 - \frac{0.0051}{0.690} \times 0.276\right) = 0.378$$

$$x_{249} = \mu + \sqrt{\beta_{250}}\, z = 0.378 + 0.071 \times 0.8 = 0.435$$

$$\text{the network's guess of the clean pixel: } \hat{x}_0 = \frac{0.379 - 0.690 \times 0.276}{0.724} = 0.26$$

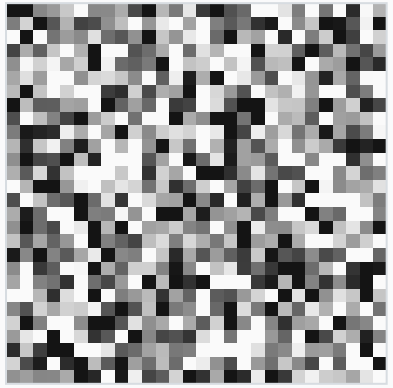
> Along the way we can read off the network's guess of the clean pixel: 0.26. Probably ink, not sure yet.

Ho, Jain & Abbeel, "Denoising Diffusion Probabilistic Models" (2020) (Algorithm 2)

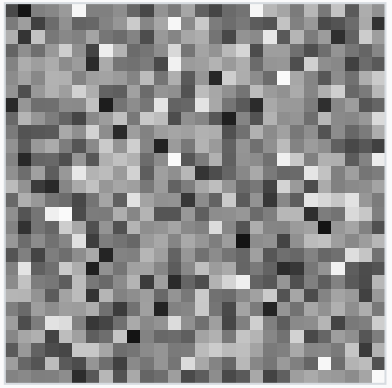
▶ Notebook 4.1

The same step on an image, at t = 600

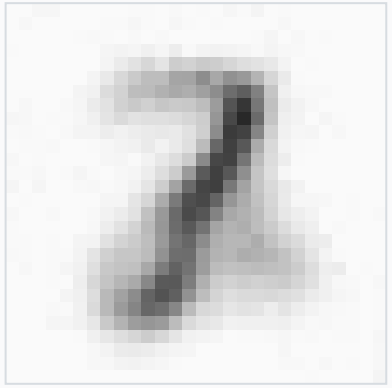
x_{600}



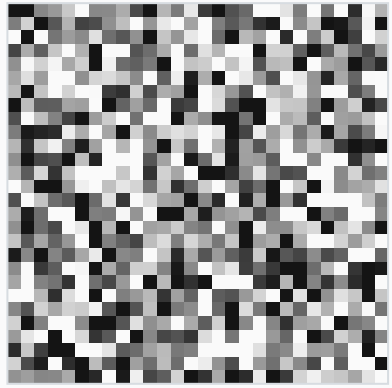
$\hat{\varepsilon}$: guessed noise



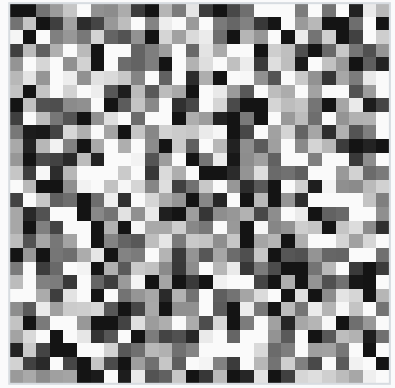
$\hat{x_0}$: guessed digit



μ: the mean



x_{599}



$\beta = 0.012$, $\bar{\alpha} = 0.026$, step width 0.110

> The mean moves the static only very slightly towards that digit, and fresh noise is added. To the eye nothing changes; over 600 more steps the 7 appears.

▶ Notebook 4.2

The toy, all the way back: every number

t	x_t	$\hat{\varepsilon}$ (network)	mean $\mu = (x_t - c_t\,\hat{\varepsilon}) / \sqrt{\alpha_t}$	fresh noise $\sqrt{\beta_t}\,z$	x_{t-1}
5	0.359	0.420	(0.359 − 0.821 × 0.420) / 0.447 = 0.033	0.894 × (−0.3)	**−0.235**
4	−0.235	−0.019	(−0.235 − 0.578 × (−0.019)) / 0.707 = −0.318	0.707 × 0.5	**0.036**
3	0.036	0.306	(0.036 − 0.426 × 0.306) / 0.837 = −0.113	0.548 × 0.2	**−0.003**
2	−0.003	0.490	(−0.003 − 0.378 × 0.490) / 0.894 = −0.211	0.447 × (−0.4)	**−0.390**
1	−0.390	1.765	(−0.390 − 0.316 × 1.765) / 0.949 = −0.999	none	**−0.999**

$c_t = \beta_t/\sqrt{(1 - \bar{\alpha}_t)}$. $\hat{\varepsilon}$ is the best possible guess for this one-pixel dataset (optional depth shows how to compute it).

> Step 1, no fresh noise: $x_0 = -0.999$. Paper. Generation does not undo our forward run: it draws a new pixel. Run it many times (next slide) and about 35% come out ink, as in the data.

▶ Notebook 4.2

Try it: walk back from noise, one step at a time

1 · Destroy | 2 · Undo | 3 · Train | **4 · Generate** | 5 · Steer | 6 · Faster | 7 · What it learns

$t = 2, \qquad x_2 = -0.003$

$\hat{\varepsilon} = \text{network}(x_2, 2) = 0.490$

the best possible guess for this one-pixel dataset

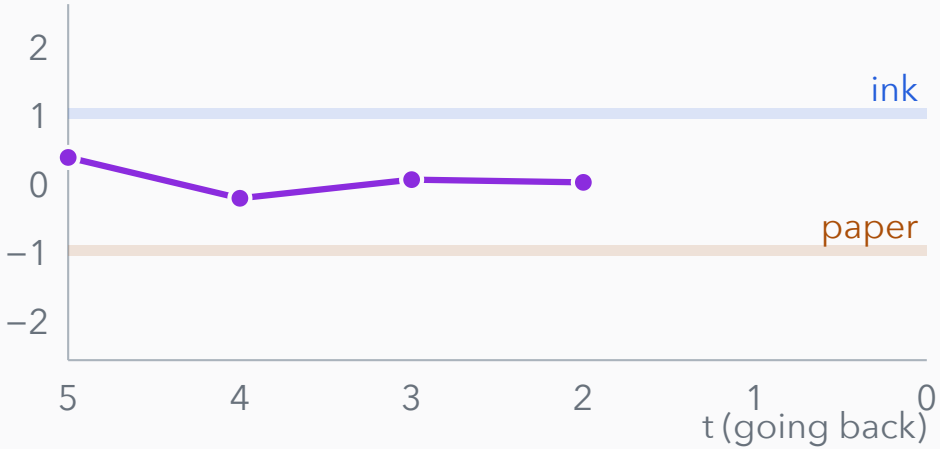
$$\mu = \frac{1}{\sqrt{\alpha_2}}\left(x_2 - \frac{\beta_2}{\sqrt{1-\bar{\alpha}_2}}\,\hat{\varepsilon}\right) = \frac{1}{0.894}\left(-0.003 - \frac{0.2}{0.529} \times 0.490\right) = -0.211$$

$$x_1 = \mu + \sqrt{\beta_2}\, z = -0.211 + 0.447\, z$$

the network's guess of the clean pixel: $\hat{x}_0 = \frac{x_2 - \sqrt{1-\bar{\alpha}_2}\,\hat{\varepsilon}}{\sqrt{\bar{\alpha}_2}} = -0.310$

previous step: $x_2 = -0.113 + 0.548 \times 0.200 = -0.003$

run 1 · finished runs: ink 0, paper 0



> Step back one click at a time and watch every number: the network's guess, the mean, the fresh noise. Start new runs and count how often you end at ink. Then switch to the real schedule: 1,000 tiny steps.

▶ Notebook 4.3

The sampling loop: start from noise, repeat a thousand times


```
x = torch.randn(n, 1, 28, 28)                # start: pure noise
for t in range(T, 0, -1):                    # 1000, 999, ..., 1
    eps_hat = model(x, t)                    # guess the noise
    c = beta[t] / (1 - alpha_bar[t]).sqrt()
    mean = (x - c * eps_hat) / alpha[t].sqrt()   # the mean of 2.4
    z = torch.randn_like(x) if t > 1 else 0  # no noise at the last step
    x = mean + sigma[t] * z                  # sample the Gaussian step
return x
```

Training uses one random level per example; **sampling** visits every level, in order: 6.1 s for 64 digits on a laptop GPU.

> Training never runs this loop; sampling always does. That is why sampling is slow, and Part 6 makes it faster.

Ho, Jain & Abbeel, "Denoising Diffusion Probabilistic Models" (2020) (Algorithm 2)

▶ Notebook 4.4

Watch 300 particles find the smiley, and drop your own

t = 0

300 particles, each its own DDPM run

orange: three of their paths

faint blue: the training data

> Press play: the network pulls them onto the face. Click anywhere in the square to drop your own particle: the network, running in this page, carries it home.

Ari Seff, "What are Diffusion Models?" (2022) (5:17) · MIT 6.S183 "A Practical Introduction to Diffusion Models" (IAP 2025), Lecture 1

▶ Notebook 4.4

Eight runs from noise: x_t, and the network's guess of the end

x_t at t = 0



$\hat{x}_0$: the network's guess of the finished digit



> Watch the fourth run: its guess changes its mind, a 0, then a 9, a 5, a 3, and finally a 9. Early steps choose the rough shape; late steps add the details.

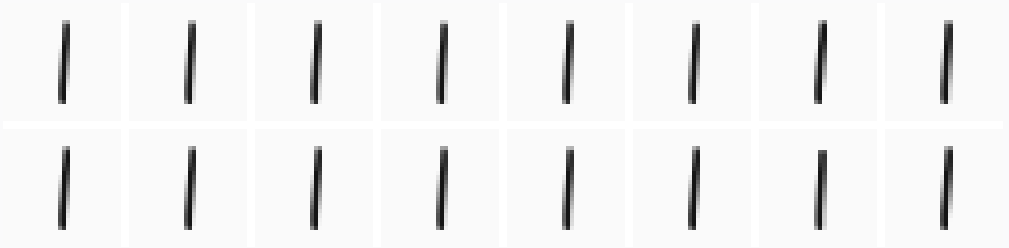
▶ Notebook 4.5

Take out the fresh noise, and every sample collapses

with fresh noise



same starts, no fresh noise



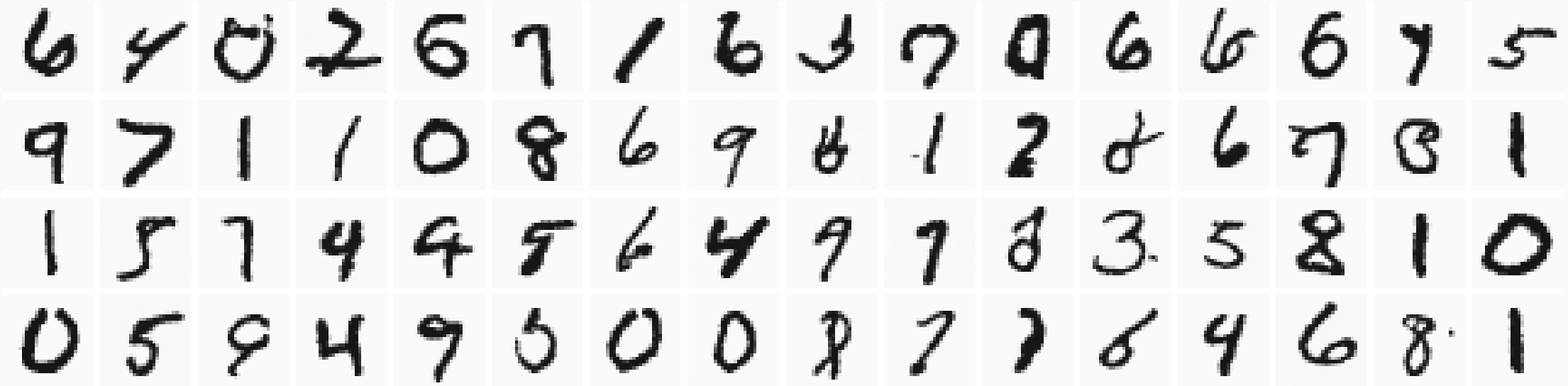
The smiley does the same: without fresh noise all 300 particles end at one point. Each mean step pulls towards an average; only the noise lets a path commit, as our pixel must commit to paper or ink.

> In 2-D every particle collapses to a single point. The step-back mean is a pull towards the average guess; the fresh noise is what keeps the samples spread like the data.

▶ Notebook 4.6

64 new digits

1 · Destroy | 2 · Undo | 3 · Train | **4 · Generate** | 5 · Steer | 6 · Faster | 7 · What it learns



Sharp like the GAN's, trained with one regression loss like the VAE. And none is a copy of a training image: Part 7 measures it.

> This is what we were after: sharp digits from a stable, simple training loss.

▶ Notebook 4.6

The whole method on a napkin

Train

1. x_0 from the data, $t \sim 1 \dots T$, $\varepsilon \sim N(0, I)$
2. $x_t = \sqrt{\bar{\alpha}_t}\, x_0 + \sqrt{(1 - \bar{\alpha}_t)}\, \varepsilon$
3. step on $\|\varepsilon - \hat{\varepsilon}_\theta(x_t, t)\|^2$

Sample

1. $x_T \sim N(0, I)$
2. for $t = T \dots 1$: $\hat{\varepsilon} = \hat{\varepsilon}_\theta(x_t, t)$
3. $x_{t-1} = (x_t - \beta_t/\sqrt{(1-\bar{\alpha}_t)}\, \hat{\varepsilon})/\sqrt{\alpha_t} + \sigma_t\, z$

End of the first session. Next: steer it with labels and text, make it faster, and see what the network really learns.

> Next session: steering with labels and text, faster sampling, and what the network really learns.

Paras Chopra, "Diffusion Models from First Principles" (2025) ("derive it on a napkin")

▶ Notebook 5.1

A 3 on demand: give the network the label


```
h = time_mlp(time_features(t))      # how noisy
h = h + label_embedding(c)            # which digit: one added line
eps_hat = unet(x_t, h)
```



Asked for a 3: **94%** are classified as 3 by a separate MNIST classifier (98% test accuracy).

Good, but some are weak or ambiguous. Can we push the samples to be more clearly 3s?

> Not all are convincing. Guidance pushes harder, and it comes from Bayes' rule.

▶ Notebook 5.2

Guidance: ask twice, then exaggerate the difference

$$\hat{\varepsilon}_{\text{guided}} = \hat{\varepsilon}(x_t, t, \varnothing) + w\left(\hat{\varepsilon}(x_t, t, c) - \hat{\varepsilon}(x_t, t, \varnothing)\right)$$

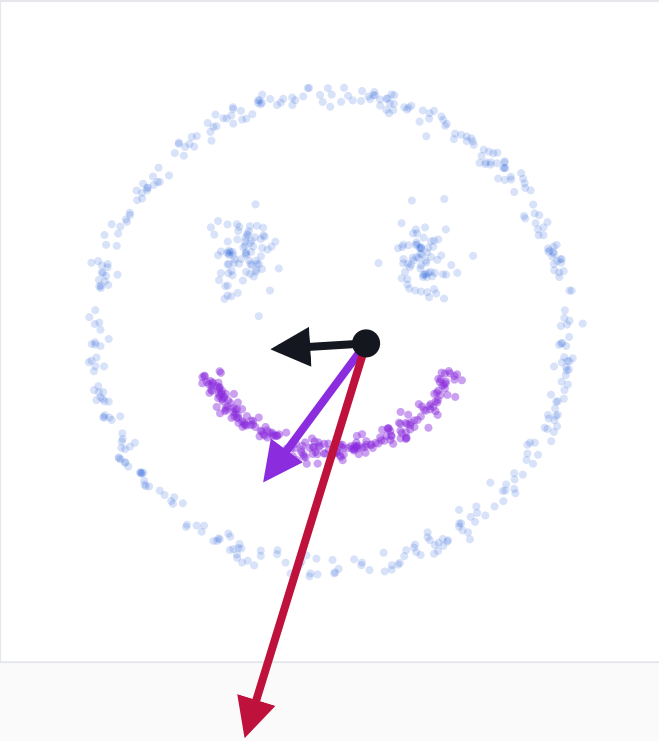
w = 0 ignores the label · w = 1 is the conditional model · w > 1 exaggerates it.

```
e_none = model(x_t, t, NULL)          # the label hidden
e_c    = model(x_t, t, c)             # the label shown
eps_hat = e_none + w * (e_c - e_none)
```

> One network gives both answers: in training, the label is replaced by “no label” 10% of the time. At sampling time: two calls per step.

▶ Notebook 5.2

Guidance as arrows: two guesses, and a push past them



at the black point, t = 300

no label: $\hat{\varepsilon}(\varnothing) = (0.16, 0.01)$

"mouth": $\hat{\varepsilon}(c) = (0.18, 0.24)$

guided: $\hat{\varepsilon}(\varnothing) + 3.0 \times (\hat{\varepsilon}(c) - \hat{\varepsilon}(\varnothing)) = (0.22, 0.70)$

w > 1: the red arrow goes past the purple one,
an extra push away from "any part of the face".

Arrows: the direction $-\hat{\varepsilon}$ in which each guess moves the point, drawn three times longer than $-\sqrt{(1-\bar{\alpha})}\,\hat{\varepsilon}$ to be visible.

Click in the square to move the point.

The part-conditional smiley network at one point. Black: its guess with no label. Purple: its guess for "mouth". Red: the guided guess. Move w: past 1, the red arrow overshoots the purple one, pushing harder towards the mouth. Click to move the point.

▶ Notebook 5.3

The guidance dial: more w, more 3, less variety

16 samples asked for a 3, w = 3



classified as 3

100%
variety (spread)

15.8
ink per image

17%

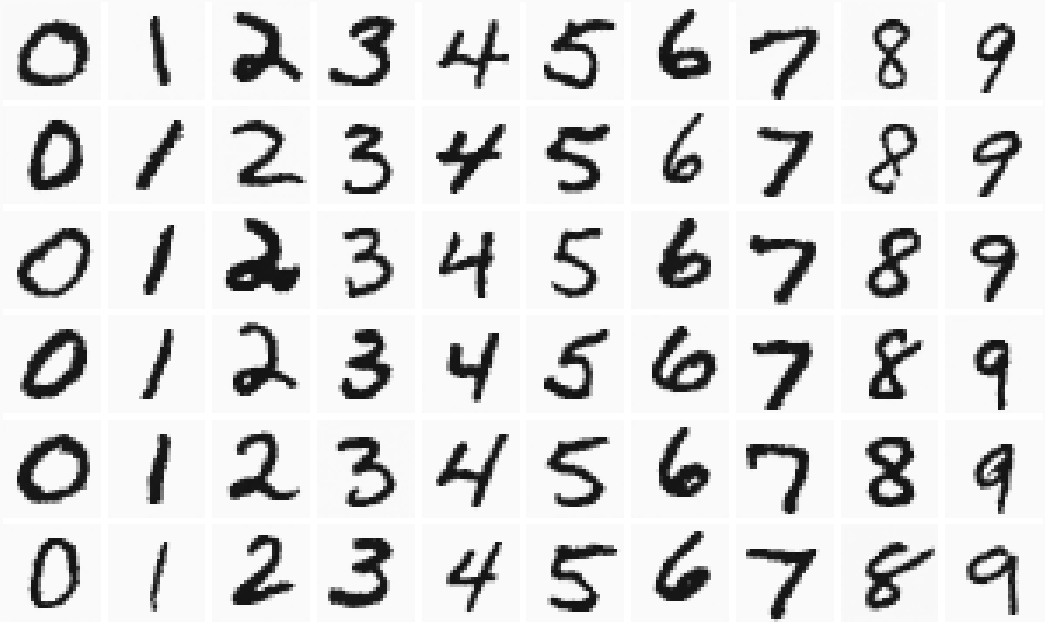
smiley guided to “mouth”, w = 3



> The smiley, guided towards the mouth: higher w pulls the samples into the middle of the mouth. Guidance trades variety for fidelity.

▶ Notebook 5.3

Every digit on demand



At w = 3 our classifier agrees with the requested digit for **100%** of these 60 samples.

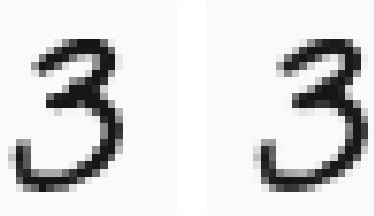
Next: scale it up, from a digit label to a sentence.

> All 60 are classified as the digit we asked for. Next: bigger images (latent diffusion), and a sentence instead of a label.

▶ Notebook 5.4

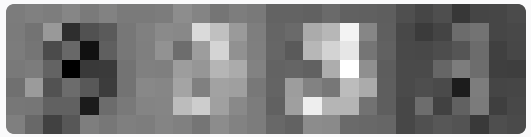
Pixels are expensive: diffuse a compressed code

A 512 × 512 colour image: **786,432** numbers through the U-Net, at every one of 50 steps, twice with guidance.



3 → 4 × 7 × 7 code (196 numbers) → decoded





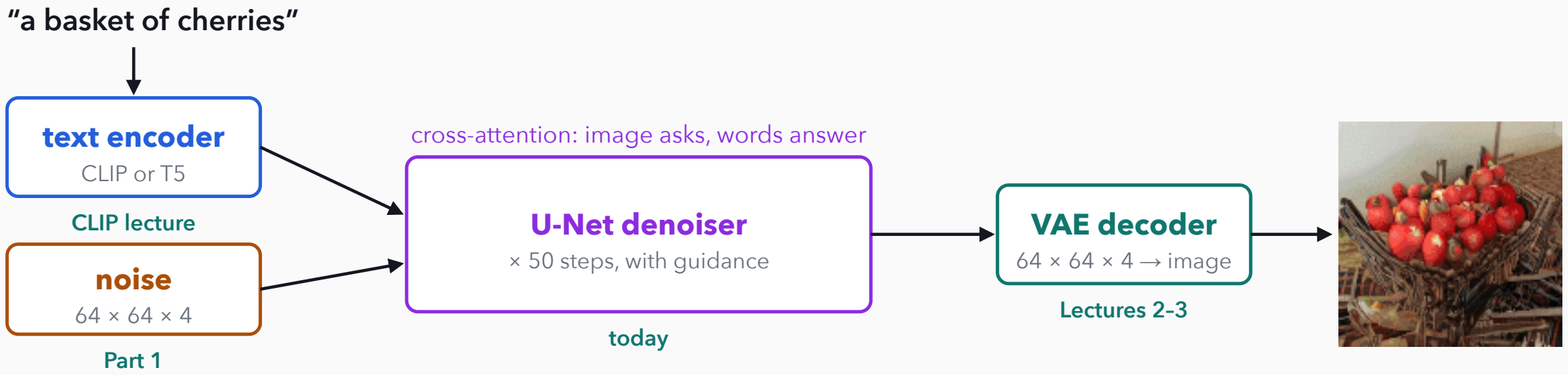
the code: four 7 × 7 channels

Diffusion on the codes, then the decoder. Stable Diffusion: 512 × 512 × 3 → 64 × 64 × 4, 48 times fewer numbers.

> Train the same diffusion model on the codes, then decode its samples: latent diffusion, the idea behind Stable Diffusion.

▶ Notebook 5.5

Text in, through cross-attention



Cross-attention: each image position asks the words a question. Queries come from the U-Net's features; keys and values from the word vectors.

> Nothing new in the training loop: image codes and captions instead of digits and labels.

▶ Notebook 6.1

A thousand network calls per image

our model, laptop GPU	time
one network call, 8 digits	5 ms
DDPM: 1,000 calls	3.2 s
a 512×512 text-to-image model	about a thousand times more work per call

Can we take fewer, bigger steps with the same trained network?

> Part 2 said big jumps back are not Gaussian. Is there another way to jump?

▶ Notebook 6.1

DDIM: jump to the guess, then re-noise it to an earlier level

$$\hat{x}_0 = \frac{x_t - \sqrt{1-\bar{\alpha}_t}\,\hat{\varepsilon}}{\sqrt{\bar{\alpha}_t}}$$

$$x_{t'} = \sqrt{\bar{\alpha}_{t'}}\,\hat{x}_0 + \sqrt{1-\bar{\alpha}_{t'}}\,\hat{\varepsilon} \qquad (t' < t)$$

$$\text{pixel, } 250 \to 200: \quad \sqrt{0.659} \times 0.26 + \sqrt{0.341} \times 0.28 = 0.373$$

> No fresh noise, so the path is deterministic, and t′ can be far earlier: 50 jumps instead of 1,000, with the same trained network.

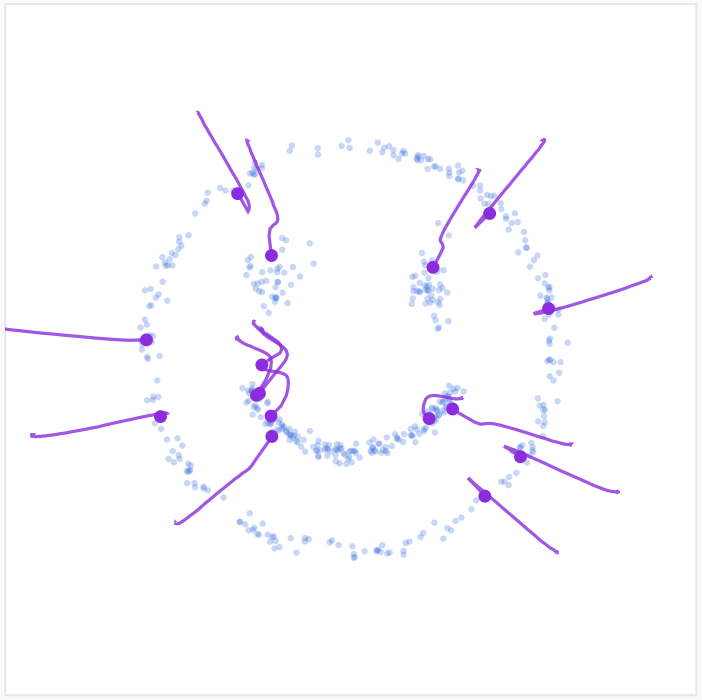
▶ Notebook 6.1

Same network: DDPM paths jitter, DDIM paths glide

DDPM: 1,000 steps with fresh noise



DDIM: 50 steps, no fresh noise

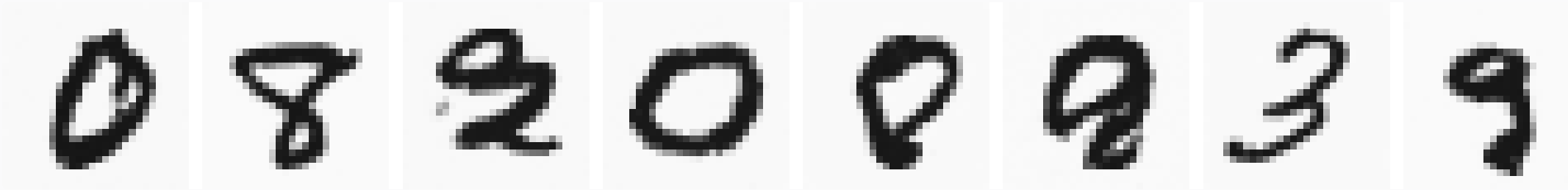


DDPM paths wander and then settle; DDIM paths are nearly straight. Both end on the smiley.

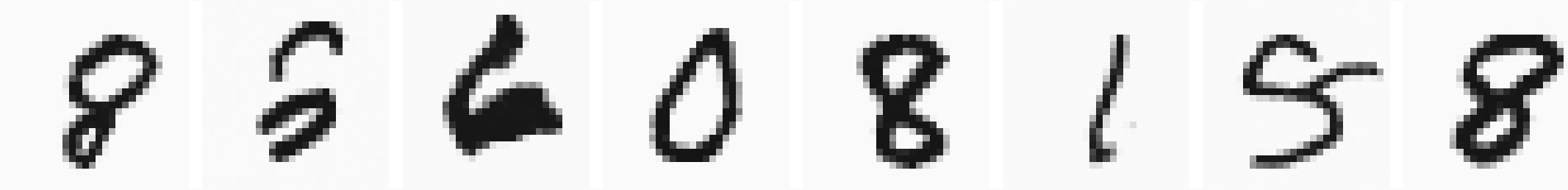
▶ Notebook 6.1

Fewer steps, same network: from 1 to 1,000

DDIM, 20 steps · 0.060 s for 8 digits



DDPM, 1,000 steps with fresh noise (same starting noise)



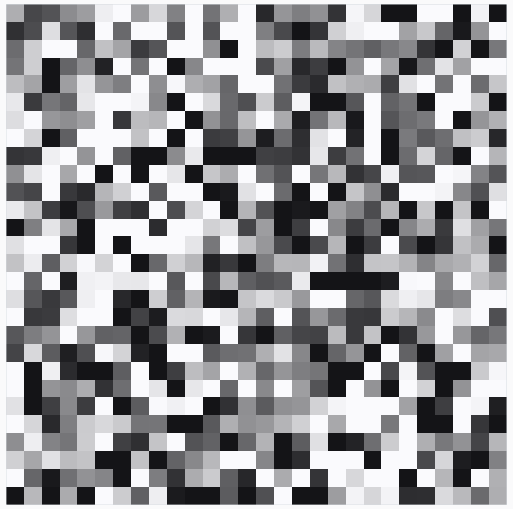
> Eight digits from the same eight noises. One step gives the blurry average of 2.2; about twenty steps already give clean digits, fifty times faster than DDPM.

Paras Chopra, "Diffusion Models from First Principles" (2025) (XI, "Speed vs Quality")

▶ Notebook 6.2

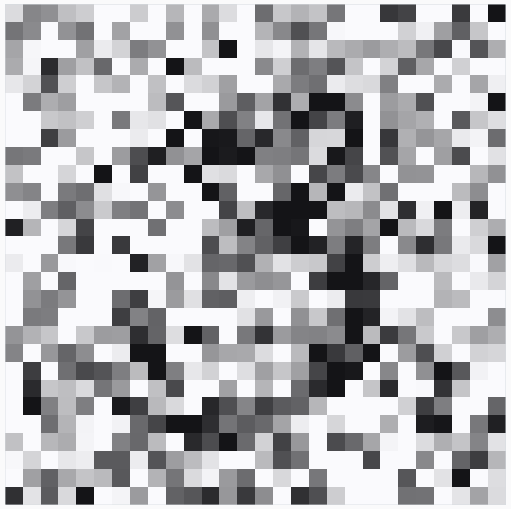
The schedule matters: linear destroys too early

linear

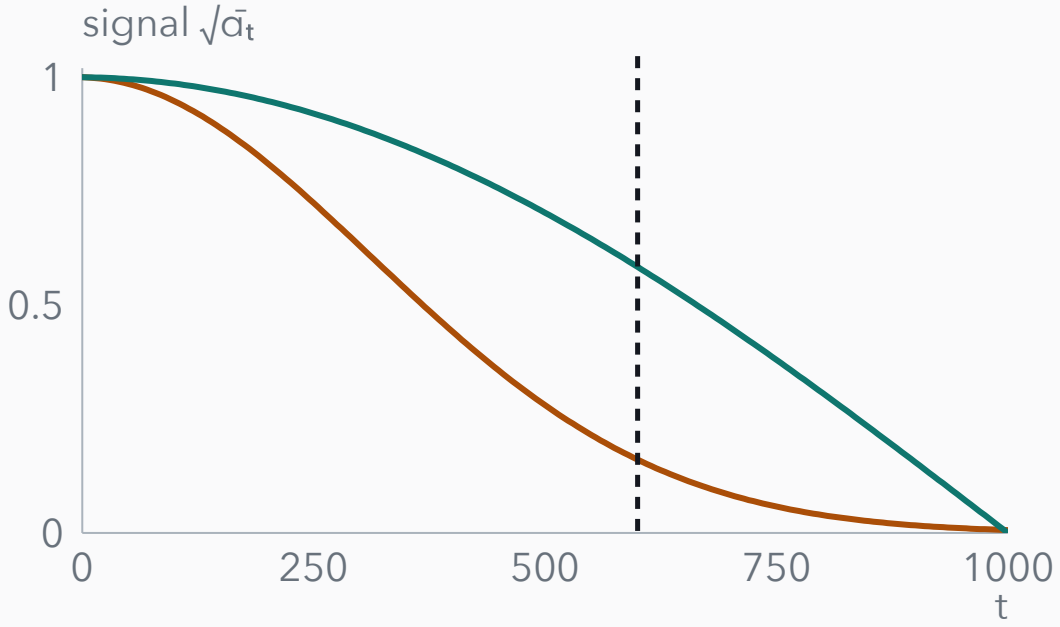


signal $\sqrt{\bar{\alpha}} = 0.161$

cosine



signal $\sqrt{\bar{\alpha}} = 0.584$



> With the linear schedule the last few hundred steps are almost pure noise, so the network spends many of its levels learning very little. The cosine schedule spreads the destruction evenly.

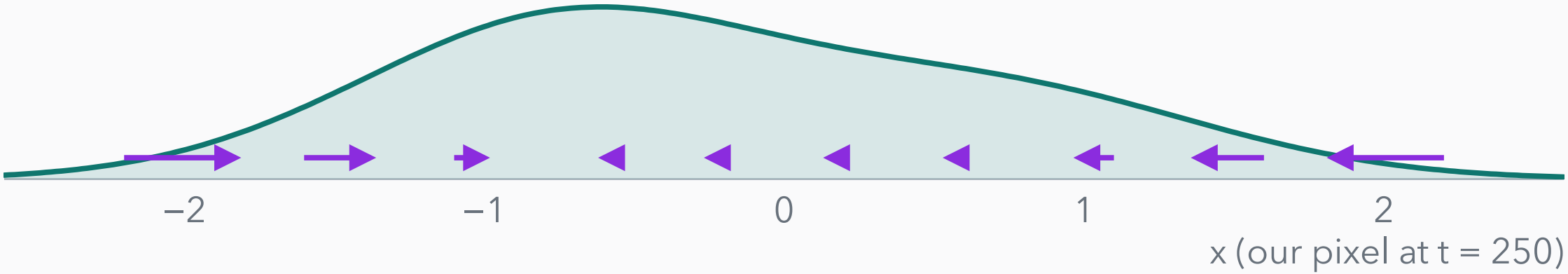
Nichol & Dhariwal, "Improved Denoising Diffusion Probabilistic Models" (2021) · Outlier, "Diffusion Models, Paper Explanation, Math Explained" (2022) (6:26)

▶ Notebook 7.1

The score: an arrow at every point, pointing uphill

$$s(x) = \nabla_x \log p(x)$$

density of the noisy pixel (green) and its score (purple arrows: uphill)

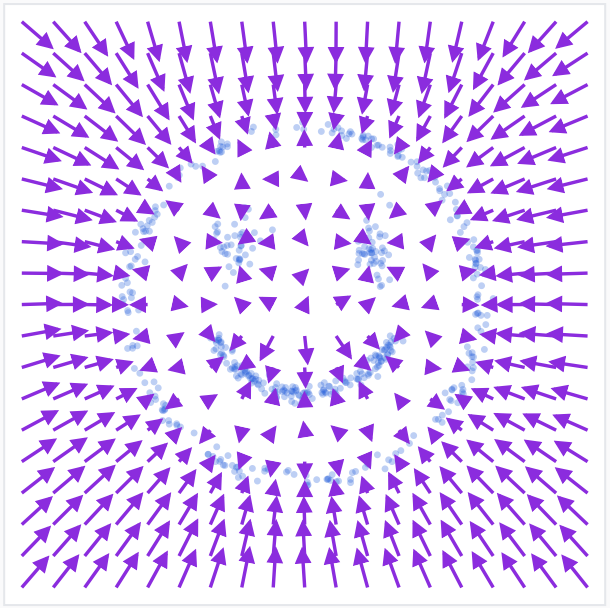


> Why it is useful: the gradient of log p ignores the normalising constant, the integral that made Lecture 1 hard. Arrows are enough to sample.

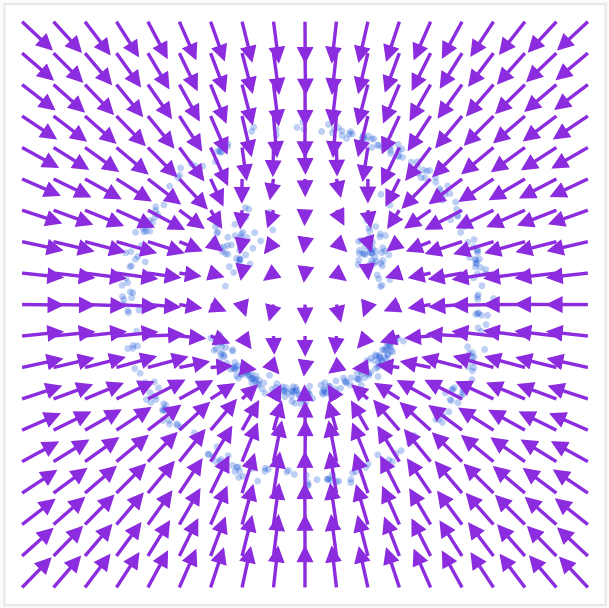
Paras Chopra, "Diffusion Models from First Principles" (2025) (IX) · Vizuara, "Principles of Diffusion Models", Lecture 1 (2025) (39:27) · Song & Ermon, "Generative Modeling by Estimating Gradients of the Data Distribution" (2019)

▶ Notebook 7.1

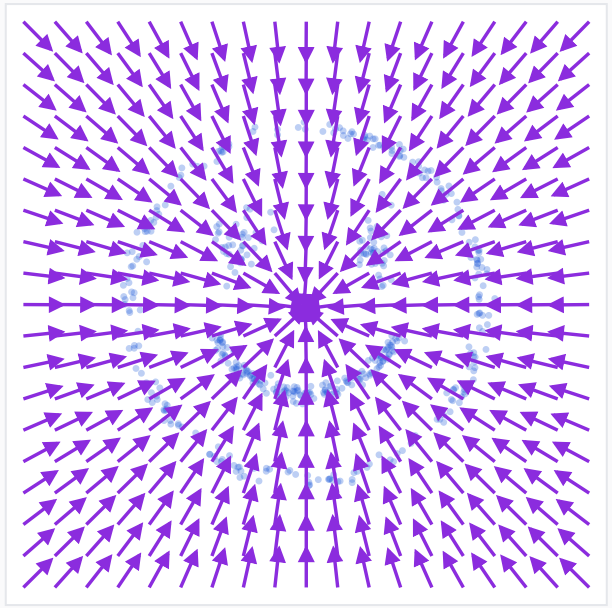
The smiley's arrows, at different noise levels



t = 10



t = 100

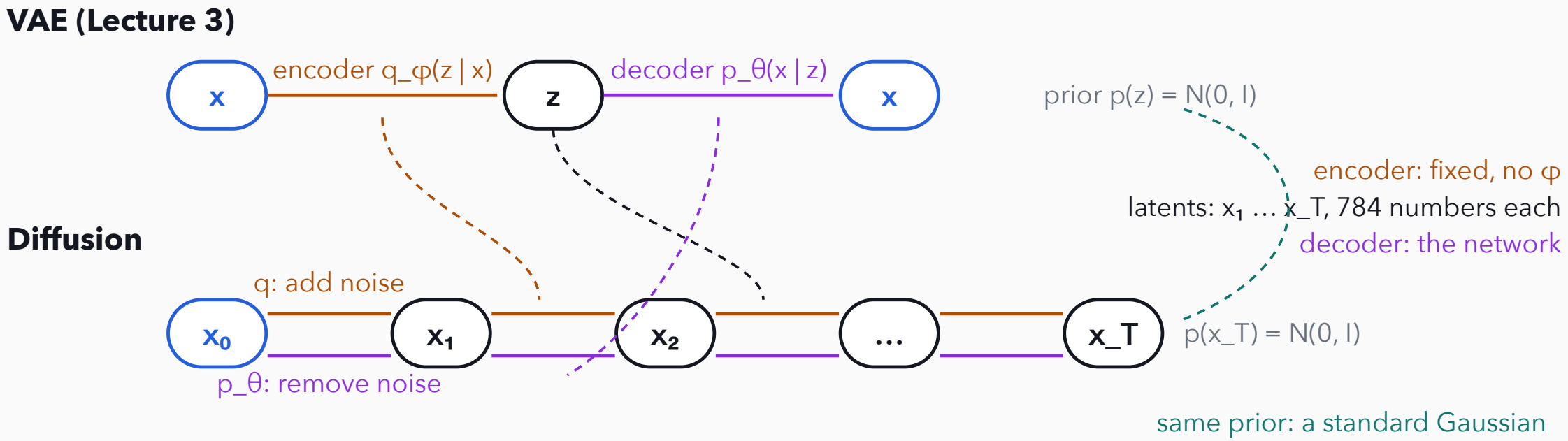


t = 500

Move t. At t = 10 the arrows are sharp near the face; far from it the network saw no training points at this level, and the arrows are less trustworthy. At t = 500 the density is blurred everywhere and the arrows are smooth.

▶ Notebook 7.2

Diffusion is a VAE with a fixed encoder



> Both start sampling from the same prior: a standard Gaussian. So we can train diffusion the way we trained the VAE: maximise the ELBO.

Ari Seff, "What are Diffusion Models?" (2022) (6:28) · Calvin Luo, "Understanding Diffusion Models: A Unified Perspective" (2022)

▶ Notebook 7.2

Its ELBO is a weighted noise error

$$-\text{ELBO} = \sum_t w_t \, \mathbb{E}\left\|\varepsilon - \hat{\varepsilon}_\theta(x_t, t)\right\|^2 + \text{const}$$

$$L_{\text{simple}} = \mathbb{E}\left\|\varepsilon - \hat{\varepsilon}_\theta(x_t, t)\right\|^2 \qquad (\text{every } w_t = 1)$$

Part 3's loss is this ELBO with its weights changed; Ho et al. found that equal weights give better samples. The derivation, one rule per line, is in optional depth.

> Set every weight to 1 and you get exactly the loss we trained in Part 3. So training is, up to this reweighting, maximum likelihood.

Ho, Jain & Abbeel, "Denoising Diffusion Probabilistic Models" (2020) · §3.4

▶ Notebook 7.3

The best possible denoiser only copies its training set

$$\hat{x}_0^*(x_t) = \sum_{i=1}^{N} w_i \, x^{(i)}, \qquad w_i \propto \exp\left(- \frac{\| x_t - \sqrt{\bar{\alpha}_t}\, x^{(i)} \|^2}{2(1-\bar{\alpha}_t)} \right)$$

sampled with our network



their nearest of 60,000 training images



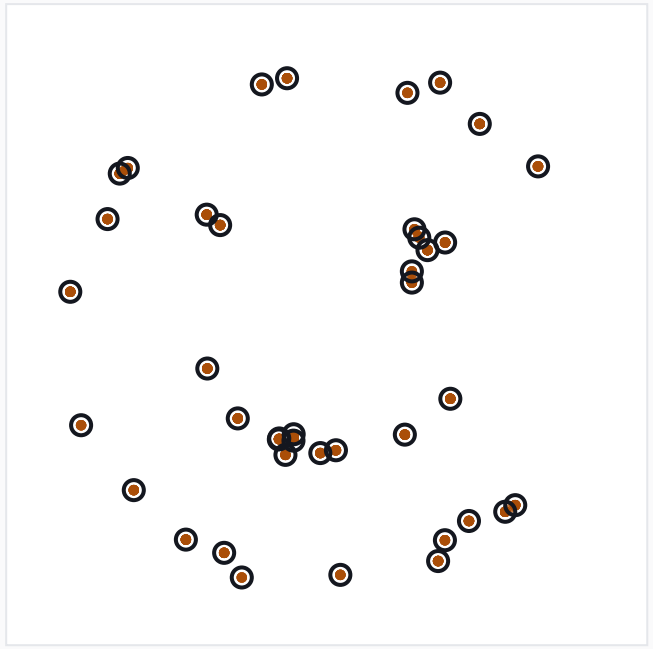
Distances **4.0–13.1**; a real test digit is 8.2 from its nearest training digit. New digits.

> Our network samples digits that are about as far from the training set as fresh test digits are. It generalises, because it cannot represent the lookup table and is smooth instead.

▶ Notebook 7.3

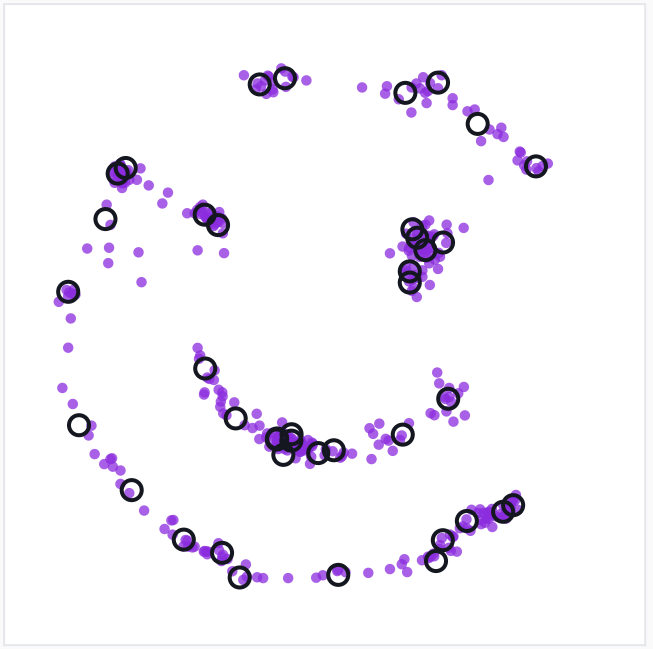
In 2-D: the perfect denoiser copies, a network fills in

the best possible denoiser



400 samples, mean distance to a training point 0.0006

our network, trained on the same 40



mean distance to a training point 0.049; to the face 0.006

> A network trained on the same 40 points: its samples spread along the curves between them, 0.049 from the nearest training point but only 0.006 from the true face. It invented the face between the points.

▶ Notebook 7.4

Where the field went: same idea, better parts

direction	what changes	examples
transformers as denoisers	the U-Net becomes a ViT on patches	DiT; Stable Diffusion 3; video models
straight paths	predict a velocity along a straight line from noise to data	flow matching; rectified flow
few steps	train a student to jump far	distillation; consistency models (1–4 steps)
beyond images	the same loop on other data	audio, video, 3-D, protein structures, robot policies, weather
text	undo masking instead of Gaussian noise	masked diffusion language models

> Since 2020 the parts changed and the idea stayed: corrupt data with a known process, learn to undo it in small steps, sample by running the steps.

MIT 6.S183 "A Practical Introduction to Diffusion Models" (IAP 2025), Lecture 2 · MIT 6.S183 "A Practical Introduction to Diffusion Models" (IAP 2025), Lecture 6 · Peebles & Xie (2023) · Lipman et al. (2023) · Liu et al. (2023) · Song et al. (2023)

▶ Notebook 7.4

Three generators, side by side

	VAE (Lecture 3)	GAN (Lecture 4)	Diffusion (today)
training signal	ELBO: reconstruct + KL	a discriminator's verdict	ELBO, reweighted: predict the noise
to sample	decode z ~ N(0, I)	generate from z ~ N(0, I)	denoise x_T ~ N(0, I), step by step
network calls per sample	1	1	20–1,000
likelihood	a lower bound	none	a lower bound
typical failure	blurry samples	unstable training, collapse	slow sampling

> The three generators of this course. Diffusion keeps the VAE's stable, likelihood-based training and gets the GAN's sharpness, and pays in sampling time.

▶ Notebook 7.4

Exit questions

1 · Destroy | 2 · Undo | 3 · Train | 4 · Generate | 5 · Steer | 6 · Faster | **7 · What it learns**

Q1 What does the network see in training, and what is its target?
x_t and t; the noise ε we drew.

Q2 Why must t be an input?
One set of weights serves every noise level; the same x_t means different things at different t.

Q3 What happens in one reverse step?
Guess $\hat{\varepsilon}$, turn it into $\hat{x}_0$, compute the mean μ, add a little fresh noise.

Q4 Why add noise while sampling?
Each step is a draw from a Gaussian; without the noise all samples collapse to one average.

Q5 Why does a perfect training loss copy the data?
The optimal denoiser of a finite set is a weighted average of training images, and sampling with it returns them.

> Q5: memorisation.

▶ Notebook 7.4

References and credits

Explainer and videos this lecture follows. Paras Chopra, "Diffusion Models from First Principles" (2025) · Ari Seff, "What are Diffusion Models?" (2022) · Deepia, "Diffusion Models: DDPM, Generative AI Animated" (2025) · Outlier, "Diffusion Models, Paper Explanation, Math Explained" (2022) · Vizuara, "Principles of Diffusion Models", Lecture 1 (2025) · MIT 6.S183 "A Practical Introduction to Diffusion Models" (IAP 2025), Lectures 1–6 (Yuan, Becker, Chen, Scarvelis, Lukoianov).

Papers. Sohl-Dickstein et al., "Deep Unsupervised Learning using Nonequilibrium Thermodynamics" (2015) · Ho, Jain & Abbeel, "Denoising Diffusion Probabilistic Models" (2020) · Song & Ermon, "Generative Modeling by Estimating Gradients of the Data Distribution" (2019) · Song et al., "Score-Based Generative Modeling through SDEs" (2021) · Song, Meng & Ermon, "Denoising Diffusion Implicit Models" (2021) · Nichol & Dhariwal, "Improved Denoising Diffusion Probabilistic Models" (2021) · Dhariwal & Nichol, "Diffusion Models Beat GANs on Image Synthesis" (2021) · Ho & Salimans, "Classifier-Free Diffusion Guidance" (2022) · Rombach et al., "High-Resolution Image Synthesis with Latent Diffusion Models" (2022) · Karras et al., "Elucidating the Design Space of Diffusion-Based Generative Models" (2022) · Lipman et al., "Flow Matching for Generative Modeling" (2023) · Liu et al., "Flow Straight and Fast" (2023) · Peebles & Xie, "Scalable Diffusion Models with Transformers" (2023) · Kadkhodaie et al., "Generalization in diffusion models arises from geometry-adaptive harmonic representations" (2024).

Tutorials. Lilian Weng, "What are Diffusion Models?" (2021) · Calvin Luo, "Understanding Diffusion Models: A Unified Perspective" (2022) · Sander Dieleman's blog (sander.ai).

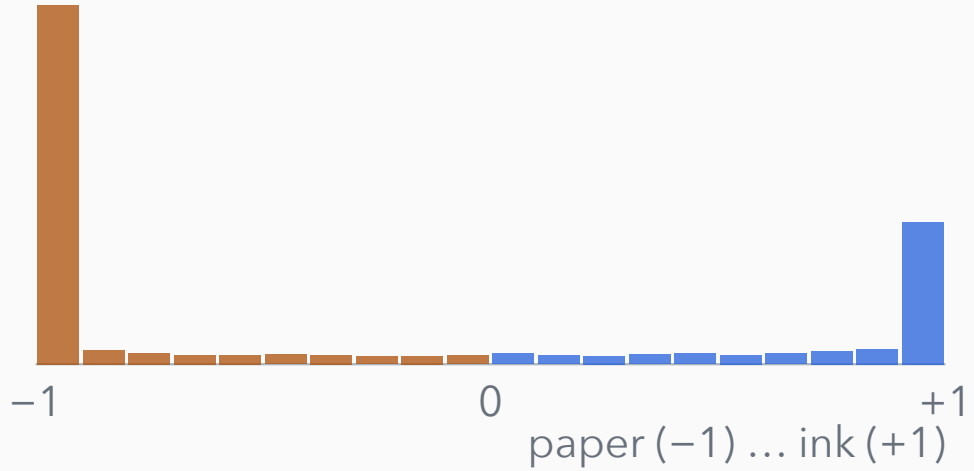
Published images (opening): LDM, RePaint, SR3, SDEdit, ControlNet, DreamBooth, each from the paper's own figures. **Illustration:** ink in water, generated with Gemini.

Our models and data. MNIST (LeCun et al.); the MNIST DDPM, conditional DDPM, autoencoder, latent DDPM and classifier from the Generative Modelling II rebuild; the smiley models trained for this lecture. Gemini (Google) watched the videos and drafted a plan; Claude (Anthropic) built the lecture.

> Every source used, with the slides that use it credited in their footers.

One pixel: paper or ink

pixel (22, 16) in 60,000 training images



Across the 60,000 training images, our pixel is ink (darker than mid-grey) in 35% of them.

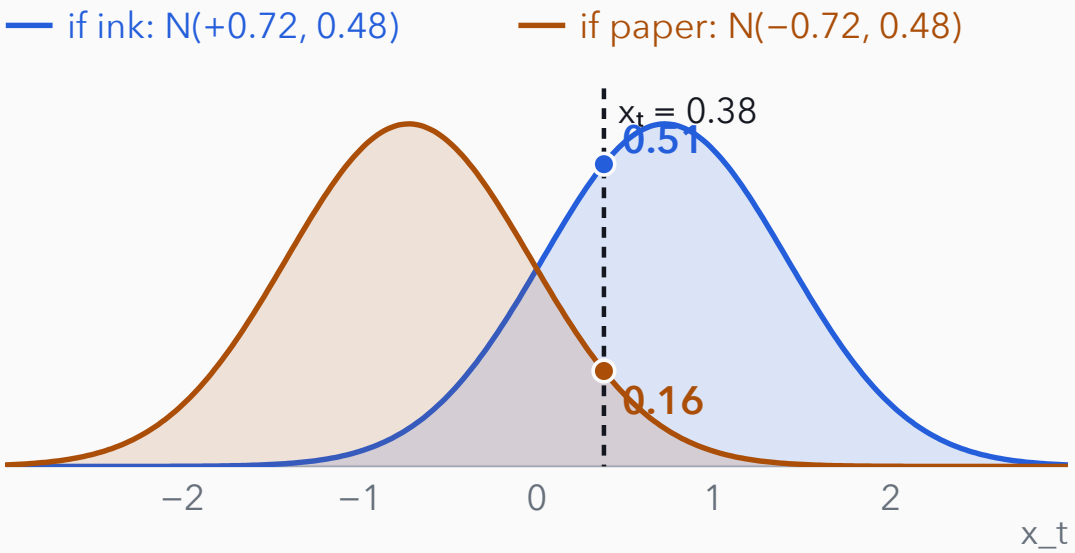
$$x_0 = +1 \text{ (ink) with probability } \pi = 0.35$$

$$x_0 = -1 \text{ (paper) otherwise}$$

A simplification: about 23% of the time this pixel is a grey in between. Ignoring the greys makes every formula exact.

> Model it as a coin: ink with probability π = 0.35, paper otherwise. This is the prior over the clean pixel, and it is all Bayes needs.

We see $x_{250} = 0.38$. Was the pixel ink?



$$p(x_{250} \mid \text{ink}) = 0.51, \qquad p(x_{250} \mid \text{paper}) = 0.16$$

$$P(\text{ink} \mid x_{250}) = \frac{0.35 \times 0.51}{0.35 \times 0.51 + 0.65 \times 0.16} = 0.63$$

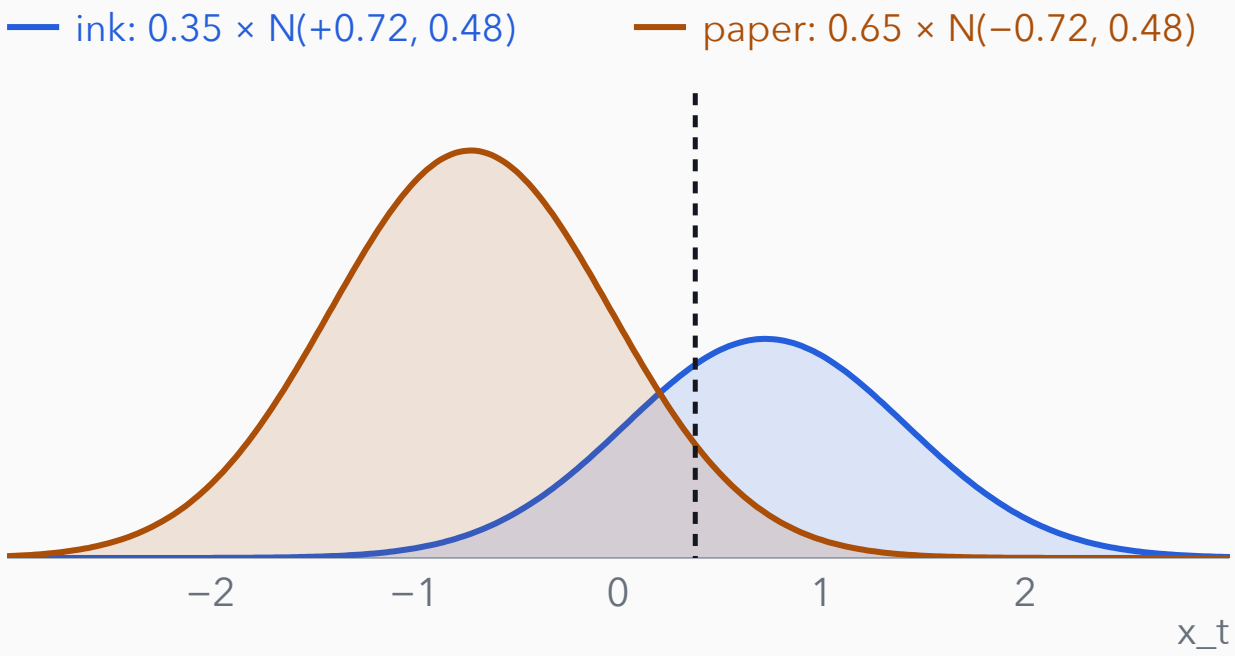
$$\hat{x}_0 = \mathbb{E}[x_0 \mid x_{250}] = 0.63\,(+1) + 0.37\,(-1) = 0.26$$

$$P(\text{ink} \mid x_t) = \sigma\!\left(\log \tfrac{\pi}{1-\pi} + \tfrac{2\sqrt{\bar{\alpha}_t}}{1-\bar{\alpha}_t}\, x_t \right)$$

In general the posterior is a sigmoid of x_t: the log of the likelihood ratio is a straight line in x_t, as in logistic regression.

▶ Notebook

Move t and x_t: the posterior and the best guess



t = 250, x_t = 0.38

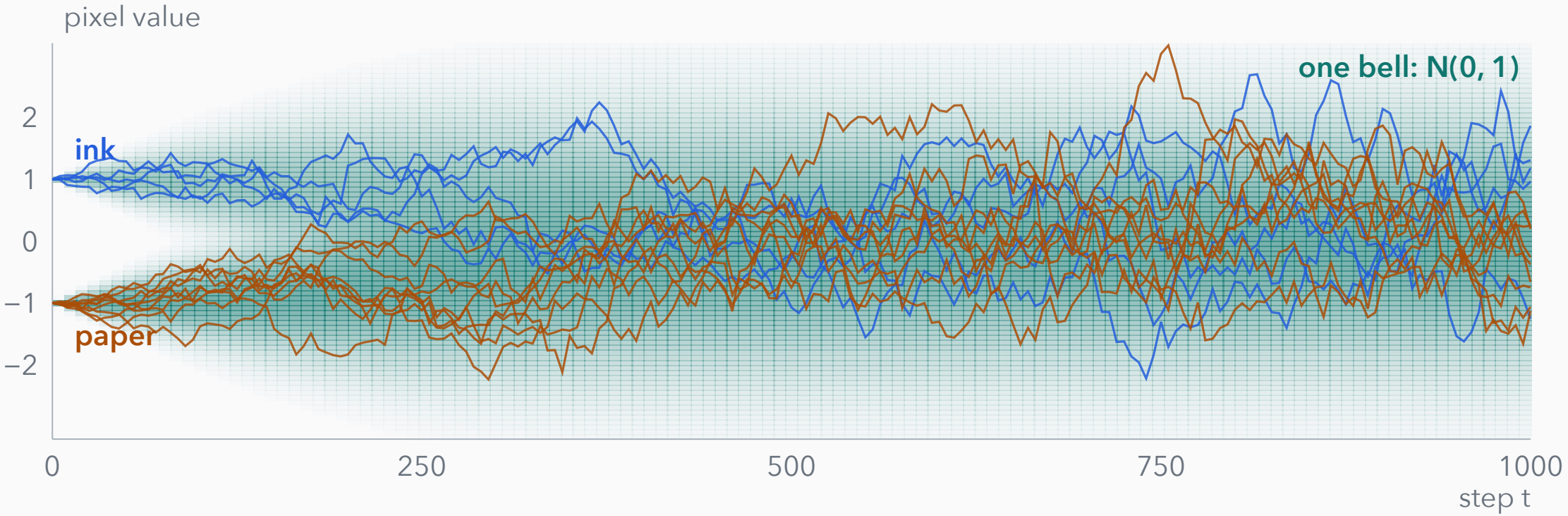
P(ink | x_t) = 0.63 paper 0.37

best guess $E[x_0 \mid x_t] = 0.26$

a grey: neither paper nor ink

> Move x_t: the posterior slides from paper to ink. Move t: at small t the two bumps are far apart and any x_t is decisive; at large t they overlap and the answer falls back to the prior.

One pixel, a thousand steps: the density flows

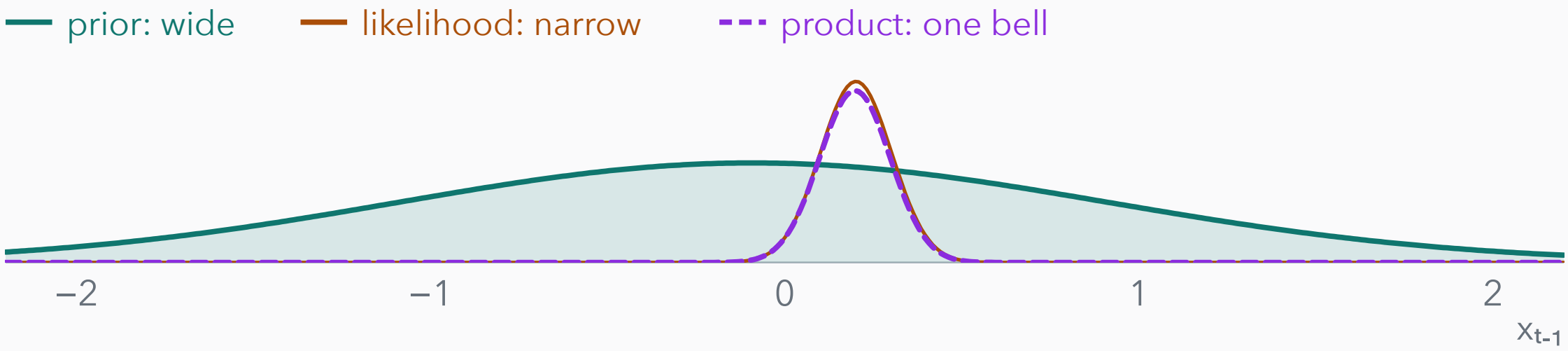


> Fourteen noising runs, step by step. Each starts at ink or paper and wanders; by the end you cannot tell where any of them started.

Deepia, “Diffusion Models: DDPM, Generative AI Animated” (2025)

Why: a narrow likelihood times a smooth prior

$$q(x_{t-1} \mid x_t) \;\propto\; \underbrace{\mathcal{N}\left(x_t;\, \sqrt{\alpha_t}\, x_{t-1},\, \beta_t\right)}_{\text{narrow: } \beta_t \le 0.02} \;\times\; \underbrace{q(x_{t-1})}_{\text{smooth}}$$



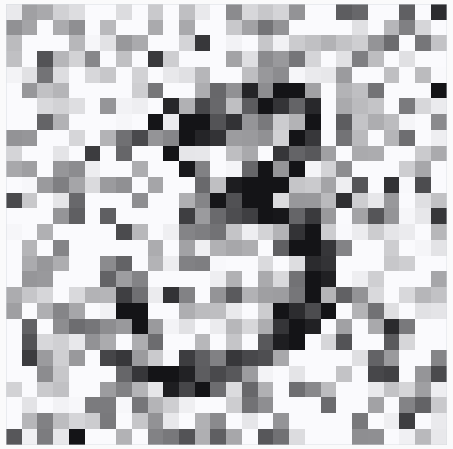
$$p_\theta(x_{t-1} \mid x_t) = \mathcal{N}\left(x_{t-1};\; \mu_\theta(x_t, t),\; \sigma_t^2\right)$$

> So we will learn one Gaussian per step: its mean comes from a network μ_θ(x_t, t); its width σ_t is fixed by the schedule. A thousand such steps, and the network is shared.

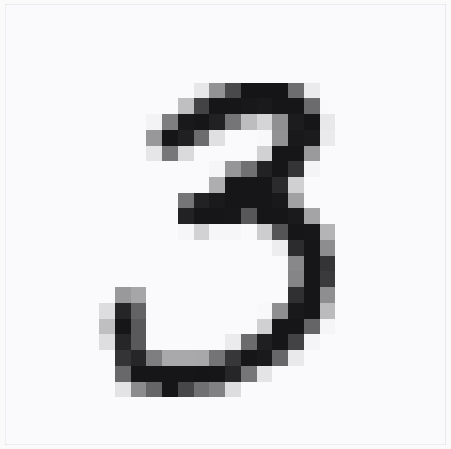
Ari Seff, "What are Diffusion Models?" (2022) · Sohl-Dickstein et al., "Deep Unsupervised Learning using Nonequilibrium Thermodynamics" (2015)

If we knew the clean image, the step back would be exact

x_{250}: noisy



x_0: known



The step back is pinned: one narrow Gaussian

pulled a little towards x_0

$$q(x_{t-1} \mid x_t, x_0) = \mathcal{N}\big(x_{t-1};\ \tilde{\mu}_t(x_t, x_0),\ \tilde{\beta}_t\big)$$

> Given the clean 3 as well, the step back is pinned down: one narrow Gaussian. During training we always have x_0, so this is a target we can compute.

Outlier, "Diffusion Models, Paper Explanation, Math Explained" (2022) (18:23)

Bayes with three Gaussians

$$q(x_{t-1} \mid x_t, x_0) = \frac{q(x_t \mid x_{t-1}, x_0)\; q(x_{t-1} \mid x_0)}{q(x_t \mid x_0)}$$

$$= \frac{\mathcal{N}\left(x_t; \sqrt{\alpha_t}\, x_{t-1}, \beta_t\right)\; \mathcal{N}\left(x_{t-1}; \sqrt{\bar{\alpha}_{t-1}}\, x_0, 1 - \bar{\alpha}_{t-1}\right)}{\mathcal{N}\left(x_t; \sqrt{\bar{\alpha}_t}\, x_0, 1 - \bar{\alpha}_t\right)}$$

$$\log q(x_{t-1} \mid x_t, x_0) = -\frac{1}{2}\left[\frac{\left(x_t - \sqrt{\alpha_t}\, x_{t-1}\right)^2}{\beta_t} + \frac{\left(x_{t-1} - \sqrt{\bar{\alpha}_{t-1}}\, x_0\right)^2}{1 - \bar{\alpha}_{t-1}}\right] + \text{const}$$

> Take logs. The denominator has no x_{t−1}: a constant. What is left is a quadratic in x_{t−1}, and an exponentiated quadratic is a Gaussian.

Outlier, "Diffusion Models, Paper Explanation, Math Explained" (2022) · Lilian Weng, "What are Diffusion Models?" (2021)

Complete the square: the mean and the width

$$\frac{1}{\tilde{\beta}_t} = \frac{\alpha_t}{\beta_t} + \frac{1}{1-\bar{\alpha}_{t-1}} \quad \Rightarrow \quad \tilde{\beta}_t = \frac{1-\bar{\alpha}_{t-1}}{1-\bar{\alpha}_t}\,\beta_t$$

$$\tilde{\mu}_t = \frac{\sqrt{\bar{\alpha}_{t-1}}\,\beta_t}{1-\bar{\alpha}_t}\,x_0 + \frac{\sqrt{\alpha_t}\,(1-\bar{\alpha}_{t-1})}{1-\bar{\alpha}_t}\,x_t$$

$$t = 250: \quad \tilde{\mu} = 0.0077 \times 1 + 0.9919 \times 0.379 = 0.384, \qquad \sqrt{\tilde{\beta}} = 0.071$$

> Our pixel: the step back keeps 99% of x_t and adds 0.8% of x_0: $\tilde{\mu} = 0.384$, with width 0.071. A tiny step.

Ho, Jain & Abbeel, "Denoising Diffusion Probabilistic Models" (2020) · eq. 6–7

We do not have x_0 when generating, so guess it

$$\mu_\theta(x_t, t) = \tilde{\mu}_t\big(x_t,\ \hat{x}_0(x_t, t)\big)$$

$$\text{pixel:} \quad \mu = 0.0077 \times 0.26 + 0.9919 \times 0.379 = 0.378 \qquad (\text{exact: } 0.384)$$

$$\hat{x}_0 = \frac{x_t - \sqrt{1 - \bar{\alpha}_t}\ \hat{\varepsilon}}{\sqrt{\bar{\alpha}_t}} \qquad (\text{the shortcut, solved for } x_0)$$

> Guessing x_0 and guessing the noise are the same thing: solve the shortcut for x_0. DDPM's network guesses the noise $\hat{\varepsilon}$.

Substitute: the mean in terms of the noise

$$\tilde{\mu}_t = a\,x_0 + b\,x_t, \qquad a = \frac{\sqrt{\bar{\alpha}_{t-1}}\,\beta_t}{1-\bar{\alpha}_t},\ \ b = \frac{\sqrt{\alpha_t}\,(1-\bar{\alpha}_{t-1})}{1-\bar{\alpha}_t}$$

$$x_0 = \frac{x_t - \sqrt{1-\bar{\alpha}_t}\,\varepsilon}{\sqrt{\bar{\alpha}_t}} \ \Rightarrow\ \tilde{\mu}_t = \left(\frac{a}{\sqrt{\bar{\alpha}_t}} + b\right)x_t - \frac{a\sqrt{1-\bar{\alpha}_t}}{\sqrt{\bar{\alpha}_t}}\,\varepsilon$$

$$\tilde{\mu}_t = \frac{1}{\sqrt{\alpha_t}}\left(x_t - \frac{\beta_t}{\sqrt{1-\bar{\alpha}_t}}\,\varepsilon\right) \qquad \frac{1}{0.9974}\left(0.379 - \frac{0.0051}{0.690}(-0.5)\right) = 0.384$$

> Simplify the two coefficients ($\sqrt{\bar{\alpha}_t} = \sqrt{\alpha_t}\,\sqrt{\bar{\alpha}_{t-1}}$ and $\beta + \alpha = 1$): $\tilde{\mu} = (x_t - \beta_t/\sqrt{(1-\bar{\alpha}_t)}\cdot\varepsilon)/\sqrt{\alpha_t}$. Our pixel with its true $\varepsilon = -0.5$: 0.384 again.

Ho, Jain & Abbeel, "Denoising Diffusion Probabilistic Models" (2020) · eq. 11

The same bound as Lecture 1

$$\text{Lecture 1:} \quad \log p(\mathcal{D}) \;\geq\; \mathbb{E}_q\big[\log p(\mathcal{D},\theta) - \log q(\theta)\big] = \text{ELBO}$$

$$\text{now:} \quad \log p_\theta(x_0) \;\geq\; \mathbb{E}_q\big[\log p_\theta(x_{0:T}) - \log q(x_{1:T} \mid x_0)\big]$$

$$p_\theta(x_{0:T}) = p(x_T)\prod_{t=1}^{T} p_\theta(x_{t-1} \mid x_t), \qquad q(x_{1:T} \mid x_0) = \prod_{t=1}^{T} q(x_t \mid x_{t-1})$$

> Both chains factorise into steps: the model's joint is the prior on x_T times a thousand learned steps back; q is a thousand fixed steps forward.

Ho, Jain & Abbeel, "Denoising Diffusion Probabilistic Models" (2020) · Lecture 1 §4.1

Turn each forward step around, given x_0

$$\mathrm{ELBO} = \mathbb{E}_q\left[\log p(x_T) + \sum_{t=1}^{T} \log \frac{p_\theta(x_{t-1} \mid x_t)}{q(x_t \mid x_{t-1})}\right]$$

$$q(x_t \mid x_{t-1}) = q(x_t \mid x_{t-1}, x_0) = \frac{q(x_{t-1} \mid x_t, x_0)\, q(x_t \mid x_0)}{q(x_{t-1} \mid x_0)} \qquad (t \geq 2)$$

$$\prod_{t=2}^{T} \frac{q(x_t \mid x_0)}{q(x_{t-1} \mid x_0)} = \frac{q(x_2 \mid x_0)}{q(x_1 \mid x_0)} \cdot \frac{q(x_3 \mid x_0)}{q(x_2 \mid x_0)} \cdots = \frac{q(x_T \mid x_0)}{q(x_1 \mid x_0)}$$

> Those ratios telescope: every middle term cancels, leaving only the two ends.

Outlier, "Diffusion Models, Paper Explanation, Math Explained" (2022) (16:23) · Calvin Luo, "Understanding Diffusion Models: A Unified Perspective" (2022)

Three kinds of term; the middle ones are the training loss

$$-\mathrm{ELBO} = \underbrace{\mathrm{KL}\big(q(x_T \mid x_0) \,\|\, p(x_T)\big)}_{L_T} + \sum_{t=2}^{T} \underbrace{\mathbb{E}_q \,\mathrm{KL}\big(q(x_{t-1} \mid x_t, x_0) \,\|\, p_\theta(x_{t-1} \mid x_t)\big)}_{L_{t-1}} \underbrace{- \mathbb{E}_q \log p_\theta(x_0 \mid x_1)}_{L_0}$$

term	what it compares	depends on θ?
L_T	the noised image with the N(0, I) prior	no: and it is ≈ 0
L_0	the very last step back	yes, a small decoder term

Each L_{t-1} compares the exact step back given x_0 (2.5) with the network's step back. For our pixel at t = 250 it is 0.0032 nats.

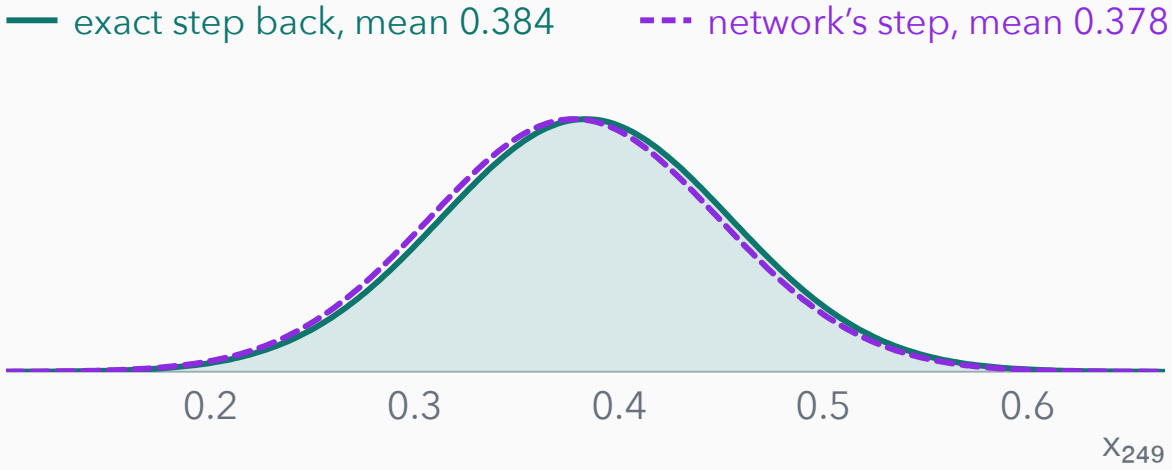
> The heart of it: a thousand KLs, each between the exact step back given x_0 and the network's step back. Both are Gaussians.

Ho, Jain & Abbeel, "Denoising Diffusion Probabilistic Models" (2020) · eq. 5

Lecture 1's formula: two Gaussians of the same width

$$\mathrm{KL}\big(\mathcal{N}(\mu_1, \sigma_1^2) \,\|\, \mathcal{N}(\mu_2, \sigma_2^2)\big) = \log \frac{\sigma_2}{\sigma_1} + \frac{\sigma_1^2 + (\mu_1 - \mu_2)^2}{2\sigma_2^2} - \frac{1}{2}$$

$$\sigma_1 = \sigma_2 = \sigma: \qquad \mathrm{KL} = 0 + \frac{\sigma^2 + (\mu_1 - \mu_2)^2}{2\sigma^2} - \frac{1}{2} = \frac{(\mu_1 - \mu_2)^2}{2\sigma^2}$$



$$\frac{(0.3836 - 0.3779)^2}{2 \times 0.00503} = 0.0032 \text{ nats}$$

> Our pixel at t = 250: the exact step back has mean 0.384, the network's (with the best guess) 0.378, both with width 0.071. KL = 0.0032 nats. Minimising it slides one bell onto the other.

Write both means with noise: the KL is a squared noise error

$$\tilde{\mu}_t = \frac{1}{\sqrt{\alpha_t}}\left(x_t - \frac{\beta_t}{\sqrt{1-\bar{\alpha}_t}}\,\varepsilon\right), \qquad \mu_\theta = \frac{1}{\sqrt{\alpha_t}}\left(x_t - \frac{\beta_t}{\sqrt{1-\bar{\alpha}_t}}\,\hat{\varepsilon}_\theta\right)$$

$$\tilde{\mu}_t - \mu_\theta = \frac{\beta_t}{\sqrt{\alpha_t}\sqrt{1-\bar{\alpha}_t}}\left(\hat{\varepsilon}_\theta - \varepsilon\right)$$

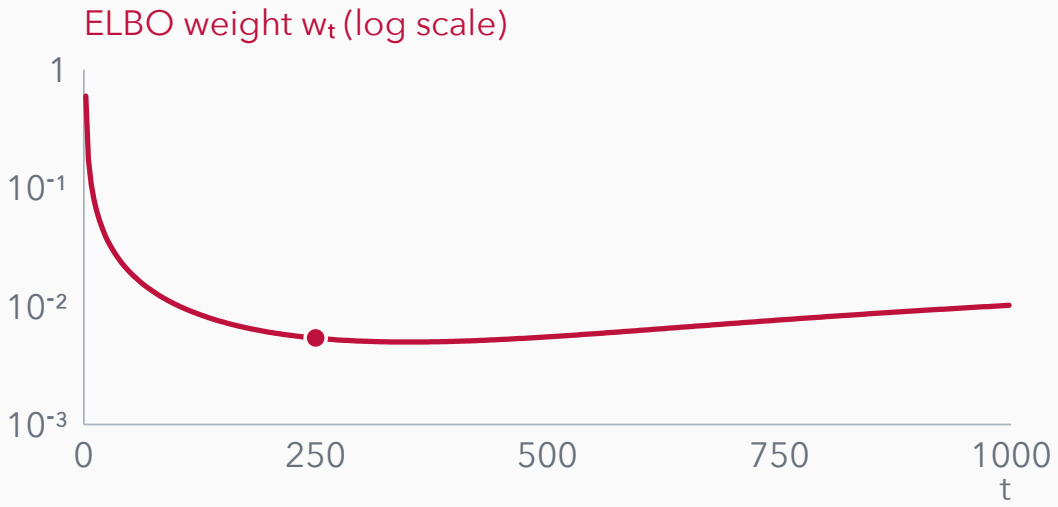
$$L_{t-1} = \frac{(\tilde{\mu}_t - \mu_\theta)^2}{2\tilde{\beta}_t} = \underbrace{\frac{\beta_t^2}{2\tilde{\beta}_t\,\alpha_t(1-\bar{\alpha}_t)}}_{w_t}\left\|\varepsilon - \hat{\varepsilon}_\theta\right\|^2$$

$$w_{250}\,(-0.5 - 0.28)^2 = 0.0054 \times 0.603 = 0.0032$$

> Square and divide by 2β̃: each KL term is a weight w_t times the squared noise error. Our pixel: 0.0054 × 0.603 = 0.0032, the same as the previous slide.

Ho, Jain & Abbeel, "Denoising Diffusion Probabilistic Models" (2020) · eq. 12

Drop the weights: the simple loss we trained in Part 3



$$L_{\text{simple}} = \mathbb{E}_{t,\, x_0,\, \varepsilon} \left\| \varepsilon - \hat{\varepsilon}_\theta(x_t, t) \right\|^2$$

Ho et al. (2020) found that equal weights give **better samples** than the true ELBO weights: they move effort from the easy small-noise steps to the harder large-noise ones.

So Part 3's loss is the ELBO with its weights changed: training is, up to this reweighting, maximum likelihood.

> DDPM drops them: every t counts equally. That is exactly the loss we trained in Part 3, now with a reason.

Ho, Jain & Abbeel, "Denoising Diffusion Probabilistic Models" (2020) · §3.4

The noise predictor is a score estimator

$$\text{one bump, } x_0 \text{ known:} \quad \nabla_x \log \mathcal{N}\left(x; \sqrt{\bar{\alpha}_t}\, x_0, 1-\bar{\alpha}_t\right) = -\frac{x - \sqrt{\bar{\alpha}_t}\, x_0}{1-\bar{\alpha}_t} = -\frac{\varepsilon}{\sqrt{1-\bar{\alpha}_t}}$$

$$\text{all bumps:} \quad \nabla_x \log q_t(x) = \mathbb{E}\left[-\frac{\varepsilon}{\sqrt{1-\bar{\alpha}_t}} \,\middle|\, x \right] = -\frac{\mathbb{E}[\varepsilon \mid x]}{\sqrt{1-\bar{\alpha}_t}}$$

$$\hat{\varepsilon}_\theta(x_t, t) \approx \mathbb{E}[\varepsilon \mid x_t] \quad \Longrightarrow \quad \hat{\varepsilon}_\theta(x_t, t) \approx -\sqrt{1-\bar{\alpha}_t}\, \nabla_x \log q_t(x_t)$$

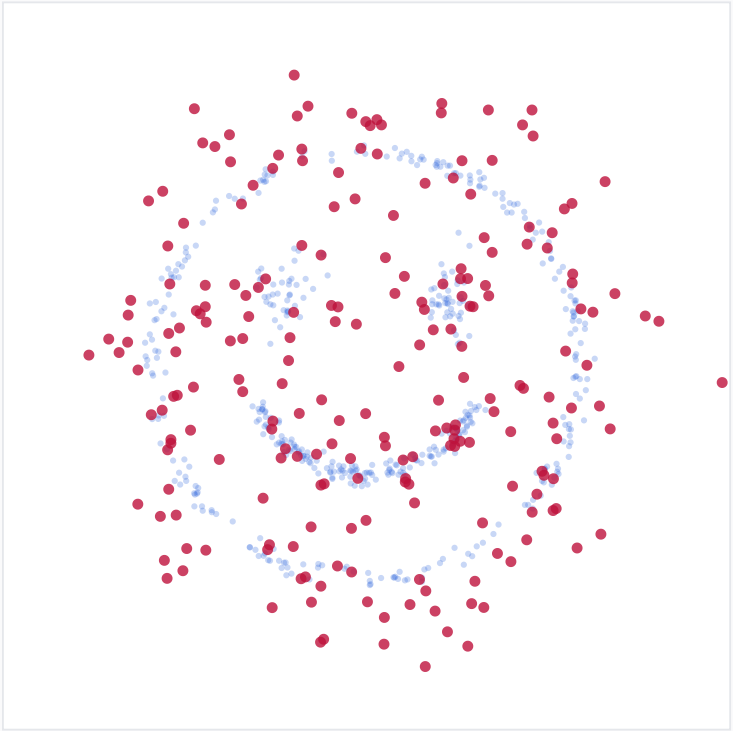
$$\text{pixel, } t = 250: \quad -\frac{-0.5}{0.690} = +0.72 \ \text{(if we knew it was ink)}, \qquad -\frac{0.276}{0.690} = -0.40$$

> Our pixel: the one-bump formula would say +0.72 (push up, towards ink) only if we knew it was ink. The true score is −0.40: it points from x_t towards $\sqrt{\bar{\alpha}} \cdot \hat{x}_0 = 0.19$.

Song & Ermon, "Generative Modeling by Estimating Gradients of the Data Distribution" (2019) · Calvin Luo, "Understanding Diffusion Models: A Unified Perspective" (2022) · Paras Chopra, "Diffusion Models from First Principles" (2025) (IX, corrected)

▶ Notebook

Follow the arrows, with a little noise: Langevin sampling



annealed, t = 1000 → 1

$x \leftarrow x + (\eta/2) \cdot \text{score}(x) + \sqrt{\eta} \cdot z$

One level: follow the arrows of that noise level.

Annealed: start at t = 1000, lower the noise step by step, as DDPM does.

> Now anneal: begin at t = 1,000 where the arrows exist everywhere, and lower the noise as the particles settle. This is what DDPM's sampler does, with a step size set by the schedule.

Four ways to run the same trained network

	random (fresh noise)	deterministic
discrete steps	DDPM: Part 4	DDIM: Part 6
continuous time	reverse SDE	probability-flow ODE

Part 5 in two sentences. Diffusion is a VAE whose encoder is fixed noising; its ELBO is a weighted sum of noise MSEs, and we train the unweighted one. The noise predictor is the score of the noisy data, so sampling climbs the arrows while the noise level falls.

> **Next:** Part 6. Use the deterministic sampler to go faster, and steer the network with a label.

MIT 6.S183 "A Practical Introduction to Diffusion Models" (IAP 2025), Lecture 2 (39:04) · Song et al., "Score-Based Generative Modeling through SDEs" (2021)

Classifier guidance: Bayes' rule on scores

$$p(x \mid c) = \frac{p(c \mid x)\, p(x)}{p(c)}$$

$$\nabla_x \log p(x \mid c) = \nabla_x \log p(x) + \nabla_x \log p(c \mid x) - \underbrace{\nabla_x \log p(c)}_{=0}$$

$$\text{guided score:} \quad \nabla_x \log p(x) + w\, \nabla_x \log p(c \mid x)$$

> Turn up the classifier's voice with a weight w > 1. The catch: we need a classifier that works on noisy images at every t.

Dhariwal & Nichol, "Diffusion Models Beat GANs on Image Synthesis" (2021) · MIT 6.S183 "A Practical Introduction to Diffusion Models" (IAP 2025), Lecture 3 (6:19)

Why guidance works: the classifier cancels out

$$\nabla_x \log p(c \mid x) = \nabla_x \log p(x \mid c) - \nabla_x \log p(x)$$

$$\nabla_x \log p(x) + w\big(\nabla_x \log p(x \mid c) - \nabla_x \log p(x)\big)$$

$$\tilde{\varepsilon} = \hat{\varepsilon}_\theta(x_t, t, \varnothing) + w\big(\hat{\varepsilon}_\theta(x_t, t, c) - \hat{\varepsilon}_\theta(x_t, t, \varnothing)\big)$$

```
e_uncond = model(x_t, t, NULL)        # label hidden
e_cond   = model(x_t, t, c)
eps_hat  = e_uncond + w * (e_cond - e_uncond)
```

> How many networks? One: during training the label is replaced by “no label” some of the time (10% for MNIST), so the same network gives both guesses. Two calls per step at sampling.

The continuous view: a stochastic differential equation

$$dx = -\tfrac{1}{2}\beta(t)\,x\,dt + \sqrt{\beta(t)}\,dW \qquad \text{(forward: the noising, with many tiny steps)}$$

$$dx = \Big[-\tfrac{1}{2}\beta(t)\,x - \beta(t)\,\nabla_x \log q_t(x)\Big]dt + \sqrt{\beta(t)}\,d\bar{W} \qquad \text{(reverse: needs the score)}$$

> Why does the reverse need the score?

Song et al., "Score-Based Generative Modeling through SDEs" (2021)

Other targets: x_0, v and flow matching

$$\hat{\varepsilon}, \quad \hat{x}_0, \quad \hat{v} = \sqrt{\bar{\alpha}_t}\,\varepsilon - \sqrt{1-\bar{\alpha}_t}\,x_0$$

$$\text{flow matching:} \quad x_t = (1-t)\,x_0 + t\,\varepsilon, \qquad \text{predict } v = \varepsilon - x_0, \quad \frac{dx}{dt} = v$$

> Why are flow-matching paths straighter?

A frozen diffusion model as a critic: score distillation

$$\nabla_\psi L_{\mathrm{SDS}} = \mathbb{E}_{t,\varepsilon}\left[w(t)\left(\hat{\varepsilon}_\theta(x_t, t, c) - \varepsilon\right)\frac{\partial x}{\partial \psi}\right], \quad x = g(\psi)$$

Render an image x from 3-D parameters ψ, noise it, ask the frozen text-to-image model for its noise guess, and push ψ so that the guess matches (DreamFusion).

> What is being trained here?